UMC UM8881/6 HB4 Super Energy Star Green chipset

Reverse Engineered

2026-02-16

Contents

Contributors **2**

Introduction **2**
- TODOs . . . 2
  - Resources . . . 3
- Chipset markings . . . 3
  - Suffix and revision analysis . . . 7
  - Serial code analysis . . . 8
- Existing PCI devices . . . 8
- Chipset address ranges . . . 8

Host Bridge (HB) UM8881-F **9**
- HB revisions . . . 9
- HB pin assignment . . . 10
  - HB schematic symbol . . . 10
  - HB pin list overview . . . 11
  - HB pin description . . . 12
- HB registers . . . 16

ISA Bridge Controller (IBC) UM8886-F/AF/BF/N **21**
- IBC suffixes . . . 21
- IBC revisions . . . 22
- IBC pin assignment . . . 22
  - IBC schematic symbol . . . 22
  - IBC UM8886-BF pin list overview . . . 25
  - IBC UM8886-BF pin description . . . 26
  - IBC UM8886-N pin list overview . . . 31
  - IBC UM8886-N pin description . . . 31
- IBC registers . . . 36
- IBC IDE controller (only AF/BF/N chip variants) . . . 39
  - IBC UM8886-AF IDE controller . . . 39
    - IBC UM8886-AF IDE config registers . . . 39
  - IBC UM8886-BF/N IDE PCI controller . . . 40
    - IBC UM8886-BF/N IDE config registers . . . 40

Super I/O (SIO) UM82C863-F, UM8663-AF/BF **41**
- SIO types and variants . . . 42
- SIO pin assignment . . . 43
  - SIO schematic symbols . . . 43
  - SIO UM82C863-F pin list overview . . . 44
  - SIO UM82C863-F pin description . . . 44

SIO UM8663-AF pin list overview . . . 47
SIO UM8663-AF pin description . . . 47
SIO registers . . . 49

UM8667 I/O coupler **51**
UM8667 I/O coupler predecessor . . . 51
UM8667 I/O coupler pin assignment . . . 51
UM8667 I/O coupler schematic symbols . . . 52
UM8667 I/O coupler pin list overview . . . 52
UM8667 I/O coupler pin description . . . 52
UM8667 I/O coupler registers . . . 54

Code snippets **54**
Fill RAM shadow segment E and F . . . 54

Contributors

Created by: pc2005 (@vogons or @retroweb discord)

Additional data directly contributed by: mkarcher (@vogons), jakethompson1 (@vogons), Landau (@retroweb), computerguy096 (@retroweb), Battler (@vogons)

Thanks to: Vogons and TheRetroWeb community

Introduction

UMC didn't make "UM8881 HB4 Super Energy Star Green" technical reference manual public, so I made my own to be able to add support into the Coreboot/SeaBIOS project. Pinouts were obtained *pains*takingly by measuring pin-by-pin on two boards:

- A-Trend ATC-1415 with a dead southbridge and NiCd damage
  - UM8881F 9536 ETA + UM8886BF 9545 DYS
  - 2x SIMM slots, single bank of up to 512 kiB L2 cache, no mouse, no VLB
- Biostar MB-8433UUD-A, a Coreboot/SeaBIOS target
  - UM8881F 9552 ETS + UM8886BF 9602 DYS
  - 4x SIMM slots, limited dual bank L2 cache, mouse, no VLB

Therefore most of the information in this document is specific to these two motherboards. Other motherboards may have a different pin mapping or use functionality not located on these two boards (especially VESA local bus).

Further the register set was checked only on MB-8433UUD-A motherboard with Am5x86-133 CPU. The functionality of some registers are too hard to decide (especially PM settings). If the chipset designed some registers to have multiple functionality (not stateless), the finding it out would be almost impossible.

So take this datasheet as always incomplete and speculative.

TODOs

Further research would be appreciated:

- Do different chipset revisions on the same board type behave differently?
- Does your cache/TAG address routing match the datasheet?
- Missing routing information for boards with 4x PCI slots (IDSEL pin, INTx routing) and routing on motherboards with VLB.
- The partial schematics of turbo button connection (or full schematics :-D)
- The meaning of SB.38 PTEST pin

- NB.170 pin should be routed to T/R (direction) pin of quad 74F245 buffers. Exact placing is yet not know (between DRAM and cache/chipset/CPU, between DRAM/cache and chipset/CPU, ...). Also it is yet not know where /OE pin of 74F245 buffers is connected to.
- DRAM refresh setting registers seems to be limited compared to other 486 chipsets (RAS before CAS, multiple timing speeds).
- What are I/O ports: `0x40b`, `0x4d6`, `0x481`, `0x482`, `0x483`, `0x487` (32bit EISA DMA?), `0xc04`, `0xc05`
- Can entire system (not only CPU) reset be generated via the software?
- PCI register space dumps for other revisions.
- SIO: Gameport pinout, pin mux support? (POST on LPT, FDC on LPT, A:/B: swap)
- SIO: What is C1.20 IDE existence bit? How does it behave?
- SIO: UM8663-BF registers/pinout and newer models

Resources

The MAIN github repository for UMC chipset. All new releases of the datasheet will be distributed here.

The MAIN Vogons discussion UMC8881/8886 Datasheet. New releases should be announced here.

The MAIN UM8881 thread on the Retro Web discord. New releases *may* be announced here if I don't forget to do that:-D.

Service manual for MiTAC notebooks, models: 5023, 5024 and 5026 or vogons thread.

DRAM pinout and naming used from wikipedia SIMM.

ISA bus pinout and naming used from wikipedia ISA.

PCI bus pinout and naming used from wikipedia Peripheral Component Interconnect.

486 pinout from 486-/PODP-Pinout and Differences. NOTICE error: A13 and A16 are switched around.

Vogons discussion about UM8881 revisions.

MSDOS PCI configuration register dumper tool Ralf Brown's PCI bus device identifier.

Linux's pciutils compiled for MSDOS.

Patents which mention "UMC Super Energy Star Green File, UM8881F/8886F": 5802555 and 5737765.

UMC IDE/EIDE controller discussion thread.

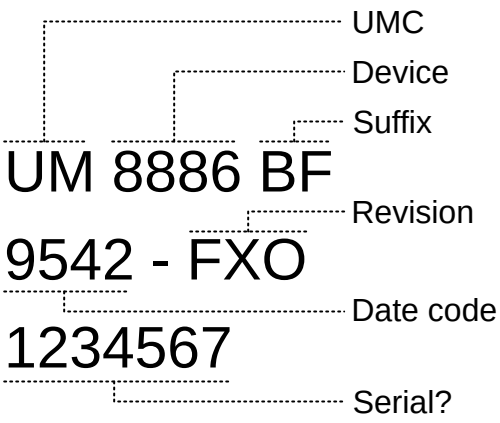
Pinouts for UMC SIO and I/O coupler discussion thread.

Discussion about adding PS/2 mouse support.

The Retro Web pages for UM82C863F SIO, UM82C865F I/O coupler (both have an independently researched pinouts), UM8663 Improved SIO, UM8667 Improved I/O coupler, UM8881F HostBridge UM8886F IBC w/o IDE, UM8886AF IBC /w ISA IDE and UM8886BF IBC /w PCI IDE.

Chipset markings

The usual marking pattern of UMC chips looks like this:



The chip functionality (pinouts, register set) depends on both suffix and revision markings. However the basic functionality seems to be the same.

A database from HB and IBC photos existing on the internet shows following combinations of these chips and the dependency on the date of manufacture.

NOTICE entries marked “?” were too blurred to be readable.

HB date-code	HB revi-sion	HB serial	IBC suf-fix	IBC date-code	IBC revi-sion	IBC serial	Motherboard
9512	BCS	R21614	AF	9514	AYS	R221E6	Aquarius Systems/BCOM MB-4DUPC
9611	ETT	137030	??				U-Board ST1A rev:B
??			AF	9510	AYS	R113E6	Soyo SY-027B2 / SY-027B5
9541	BTA	131510	AF	9527	BYA	R417B0	Shuttle HOT-433
9524	BTA	128910	AF	9544	FYO	MA03B0	QDI P4U880A/IO
9523	BCS	M42926	AF	9544	FYS	MA05C5	QDI P4U880A/IO
9521	BCA	N42649	AF	9521	AYS	M411D6	Protech PM486PU
9511	BCT	R20613	AF	9511	AYS	R205SA	PowerTech MB468
9535	ETA	133570	AF	9523	AYO	N429B6	Pine Technology PT-432A
9528	BTA	129620	AF	9523	AYA	N421F1	Pine Technology PT-431A
9548	ETS	134990	BF	9611	FXS	704050	PCPartner UMC Lite 486
9518	BCA	N30618	AF	9517	AYS	R314F1	PCPartner UMC Lite 486
95??	BTS	??	AF	9515	AYA?	??	PCPartner M486UPI
9606	ETS	134880	BF	9611	FXS	704050	PCChips M921
9535	ETO	133660	AF	9450	AYS	RA2548	PCChips M921
9524	BTO	128630	AF	9540	BYS	R525E6	PCChips M921
9540	BTO	131280	AF	9540	BYS	R603A6	PCChips M921
9524	DYA	R42526	AF	9540	FYA	N904E0	MiTAC/Trigon PL4600C
9519	DYS	R40460	AF	9525	AYS	N508B6	MiTAC/Trigon PH4500AU
9524	BTA	124300	AF	9525	AYS	N516J6	Gigabyte GA-486AM
9523	BCS	N42644	AF	9522	BYS	R423T6	Gigabyte GA-486AM
9520	BCO	R30703	AF	9519	AYS	N409D0	Gigabyte GA-486AM
9512	BCS	R11746	AF	9513	AYS	R222A7	Chicony CH-880C
9514	??	N22820	AF	9517	AYS	R329A6	Biostar MB-84xx-U
9515	BCA	N30513	AF	9514	AYA	R226F6	Biostar MB-8425UUC-A
9511	BCO	R20730	AF	9512	AYS	R215C6	Biostar MB-8425UUC
9620	ETT	13536B	BF	9618	FXS	KM3Z58	U-Board JK-042E
9631	EYA	R62913	BF	9633	FXA	KM4U53	U-Board JK-042E
9611	ETT	137030	??				U-Board JK-042E
9546	ETS	134640	BF	9547	DYS	RA26B5	A-Trend ATC-1415
9519	BCT	N32406	AF	9524	AYA	N428D0	A-Trend ATC-1415
9546	ETS	134350	BF	9547	DYS	RA25F0	A-Trend ATC-1415
9536	ETA	133680	BF	9545	DYS	R702B5	A-Trend ATC-1415

HB date-code	HB revision	HB serial	IBC suffix	IBC date-code	IBC revision	IBC serial	Motherboard
9631	EYS	R62921	BF	9625	FXA	KM4M62	Biostar MB-8425UUD-A
9634	EYT	R63009	BF	9634	FXS	RM5366	Biostar MB-8425UUD-A
9540	ETA	134740	BF	9545	DYS	R630B0	Biostar MB-8425UUD-A
9552	ETS	134630	BF	9602	DYS	RC02J0	Biostar MB-8425UUD-A
9609	ETO	136550	BF	9602	DYS	RB29C5	ECS UP8812 AIO
9645	EYS	MA0508	BF	9645	FXO	KM5T65	Ford Lian 9885AW
9506	BCS	RC2031	AF	9516	AYS	R320C1	MECER CORPORATION B885
9507	BCA	RC2708	??				MG PCI/ISA-UD 486 GRN
9549	BTA	130770	BF	9552	DYS	RB01E0	PCChips M915i
9524	BTS	129180	AF	9524	AYA	N511B6	PCChips M915i
9621	EYT	R42558	BF	9619	CYA	M322N0	PCChips M919 v1.x
9621	EYT	R42436	BF	9617	FXT	ZM3452	PCChips M919 VER 3.3B/F
9630	ETA	R62704	BF	9622	FXS	KM3Z62	PCChips M919 VER 3.4B/F
9623	EYT	R42841	BF	9622	FXT	KM4958	PCChips M919 VER 3.4B/F
9621	EYT	R42436	BF	9619	FXA	704060	PCChips M919 VER 3.4B/F
9642	EYT	M92205	BF	9641	FXA	KM5J73	Pine Technology PT-432B
9507	BCA	RC2033	AF	9509	AYS	RC24B6	Shuttle HOT-433 (Ver. 4.0)
9603	ETS	136560	BF	9602	DYS	RB15D0	Shuttle HOT-433 (Ver. 4.0)
9548	ETS	135830	BF	9604	DYS	SB29A5	Shuttle HOT-433 (Ver. 4.0)
9640	EYO	R80728	BF	9641	FXA	KM5Y53	Shuttle HOT-433 (Ver. 4.0)
9514	BCA	N30349	AF	9514	AYS	R301D1	Shuttle HOT-433 (REV. 1)
9623	EYT	R51027	BF	9636	FXA	RM5666	Sowah Research SR-M401-A
9540	ETA	134740	??				Sowah Research SR-M401-A
9651	EYA	MB1231	BF	9652	FXS	MMBM69	Sunnylab SYL8884PCI-EIC
9606	ETS	134880	BF	9611	FXS	704050	Trang Bow/Trangg Bow TB-486-OB
9614	ETT	??	BF	9615	FXT	KM3453	Trang Bow/Trangg Bow TB-486-OB
9622	EYT	R50934	AF?	9544	FYT	R911A0	Trang Bow/Trangg Bow TB-486-OB
9441	BA	R90519	F	9444	AS	R91328	ECS UM8810P-AIO
9444	BS	R91443	F	9445	BS	R93046	PowerTech MB467
9429	BS	R62582	F	9432	AS	R70166	QDI MP4-P4U880GRN V2
9511	BCT	R20613	F	9513	EYS	R11345	QDI MP4-P4U885G
9525	EYS	R62???	F	9527	EYS	??	Siemens Nixdorf System Board D882
9506	BCA	RC0620	F	9453	BS	RA0650	TK TK8880F/2066
9512	BCT	N21917	F	9513	EYS	R11345	TK TK8880F/2066A2
9636	EYT	R80720	F	9638	EYT	Z61550	AST Advantage! 575
9437	BA	R73013	F	9437	AA	R80235	Chaintech 4UPB
9435	BS	R72240	F	9437	AA	R80234	DataExpert EXP8046
9446	BCS	RA0844	F	9448	BS	RA2944	ECS UM8810P-AIO (rev 1.x)
9525	BTT	129700	F	9529	EYO	N61265	ECS UM8810P-AIO (rev 2.x)
9451	BCS	RA0433	F	9450	BS	RA3034	ECS UM8810P-AIO (rev 2.x)
9435	BS	R72336	F	9437	AA	R80540	Gigabyte GA-486IM
9448	BCA	RA2749	F	9445	BS	R92941	Gigabyte GA-486IM
9441	BS	R82721	F	9432	AS	R70569	Gigabyte GA-486IM
9449	BCS	RA0631	F	9448	BS	RA1344	Gigabyte GA-486IM
9435	BS	R72636	F	9437	AS	R73138	Gigabyte GA-486IM
9603	ETS	136100	BF	9552	DYS	RB15E0	Shuttle HOT-433 (Ver. 4.0)
9524	BTS	124020	AF	9544	FYT	MA03B5	QDI P4U880A/IO
9528	EYS	R52816	F	9526	EYS	N51484	Siemens Nixdorf System Board D882
9514	BCA	N22726	F	9513	EYS	RC0549	QDI MP4-P4U885G
9520	BCO	N30946	F	9520	EYS	R40554	QDI MP4-P4U885G
9519	BCS	N32215	AF	9519	AYS	N405M6	Pine Technology PT-431A
9620	EYS	R41235	BF	9618	FXA	RM3153	PCChips M919
9444	BS	R91445	F	9445	BS	R92841	PowerTech MB467

HB date-code	HB revi-sion	HB serial	IBC suf-fix	IBC date-code	IBC revi-sion	IBC serial	Motherboard
9544	ETA	133820	BF	9607	DYS	RB28D0	Sunnylab SYL8884PCI-EIC
9646	EYT	MA1815	BF	9651	FXS	MMBN60	PCChips M919
9540	ETA	133760	BF	9540	DYS	R618F6	Biostar MB-8433UUD-A
9512	BCT	N21918	AF	9516	AYS	R311M6	Shuttle HOT-433 (Ver. 1-3)
9434	BS	R72239	F	9434	AS	R70367	ECS UM8810P-AIO (rev 1.x)
9448	BCA	RA1747	F	9452	BS	RA2449	ECS UM8810P-AIO (rev 1.x)
9448	BCA	?A0845	F	9448	BS	SA2347	ECS UM8810P-AIO (rev 1.x)
9508	BCS	R10338	??				ECS UM8810P-AIO
9535	ETA	133570	BF	9539	DYS	R607K6	Biostar MB-8425UUD-A
9606	ETS	135370	BF	9611	FXS	615040	Biostar MB-8425UUD-A
9435	BS	R72240	F	9438	AS	R80424	ECS UM8810P-AIO (rev 1.x)
9548	ETS	136680	BF	9552	DYS	RB16E0	Biostar MB-8425UUD-A
9519	BCS	N30514	F	9519	EYS	R40159	ECS UM8810P-AIO (rev 2.x)
9552	ETS	134250	BF	9552	DYS	RB16E0	Octek HIPPO 15
9514	BCS	N30111	AF	9513	AYS	R222B1	Shuttle HOT-433 (REV. 1)
9645	EYA	MA0508	BF	9643	FXA	KM5E68	PCChips M919 VER 3.4B/F
9642	EYT	M92201	BF	9641	FXA	KM5J71	PCChips M919
9620	EYS	R41235	??stick	96??	??	RM??	PCChips M919
9640	EYO	R80728	BF	9639	FXT	KM5R57	Shuttle HOT-433 (Ver. 4.0)
9447	BCS	RA0434	F	9445	BS	RA0152	DataExpert EXP8046
9615	ETA	13686B	BF	9615	CYS	Z306E5	PCChips M919 VER 3.2
9435	BS	???	???	9435	AS	???	PCChips M915 V1.3, chipset rebranded
9651	EYA	MB1330	BF	9652	FXS	MMBN53	PCChips M919 VER 3.4B/F
9629	EYA	R51712	BF	9625	FXA	KM4960	PCChips M919 VER 3.4B/F
9512	BCS	R21614	AF	9512	AYT	R213B6	PowerTech MB468
9550	BTT	132210	BF	9549	DYS	RA25E0	PCChips/Hsin Tech M921
9544	BTA	132460	AF	9540	BYS	R530D5	Gigabyte GA-486AM
9614	ETA	134770	??stick	??	??	ZM3152(?	PCChips/Hsin Tech M921
9517	BTS	123820	F	9519	EYS	R40552	ECS UM8810P-AIO (rev 2.x)
9516	BTO	125100	AF	9517	AYS	R321E6	Gigabyte GA-486AM
9437	BA	R72812	F ?? blurry	??	??	??	Gigabyte GA-486IM
9516	BTO	124320	?? blurry	??	??	??	Gigabyte GA-486AM
9434	BS	R72239	F	9433	AS	R63068	Gigabyte GA-486IM
9602	ETS	133670	??stick	??	??	??	Sowah Research SR-M401-A
9524	BTA	128490	AF	9525	BYS	R419E1	Gigabyte GA-486AM
9435	BS	R72241	F	9442	AS	R91326	ECS UM8810P-AIO
9642	EYT	M92507	BF	9643	FXT	MM5Q58	PC Chips M919
9614	ETA	134900	BF	9616	FXS	ZM3152	PC Chips M919
9622	EYA	R42843	F	9617	EYS	MB0827	AST Advantage! 575
9546	ETS	134750	BF	9546	DYS	R618G1	Biostar MB-8433UUD-A rev 2
9633	EYT	R63012	BF	9634	FXS	RM5366	Biostar MB-8433UUD-A rev 3
9523	BCT	N42462	AF	9523	AYA	N426A0	Shuttle HOT-433
9437	BA	R73012	F	9437	AS	R73137	ECS UM8810P-AIO
9523	BCT	N42647	AF	9523	AYA	N426A0	Shuttle HOT-433
9514	BCS	N30112	AF	9514	AYA	R302D1	Shuttle HOT-433
9511	BCT	R20613	AF	9517	AYS	R328N6	Shuttle HOT-433
9523	BCT	N42462	AF	9520	AYS	R404M0	Shuttle HOT-433
9520	BCS	N32216	F	9520	EYS	R40649	TK TK8880F/2066A2
9714	EYS	EC0521	BF	9652	FXS	MMBN57	PC Chips M919 (youngest NB!, MAME)
9650	EYS	MB1227	BF	9649	FXA	MMB575	PCChips M921

HB date-code	HB revi-sion	HB serial	IBC suf-fix	IBC date-code	IBC revi-sion	IBC serial	Motherboard
9506	BCA	RC0619	F	9510	EYS	RC0344	QDI MP4-P4U885G
9514	BCS	N30620	AF	9515	AYS	R311N1	Gigabyte GA486AM/S
9523	BCT	N43037	AF	9520	BYS	R331C0	Gigabyte GA486AM/S
9634	EYT	R62916	BF	9633	FXA	KM4U53	PC Chips M919
9636	EYT	??0725	BF	9640	FXA	KM5E69	PC Chips M919
9622	EYS	R42559	BF	9618	FXS	M33640	PC Chips M919
9618	ETA	136078	BF	9611	FXS	630040	PC Chips M919
9610	ETO	136090	BF	9615	CYA	Z303A0	PC Chips M919
9618	ETA	13566B	BF	9616	CYT	M320G5	PC Chips M919
9622	EYT	R42641	BF	9622	FXA	KM4161	PC Chips M919
9514	BCS	N22816	AF	9516	AYS	R314G1	Biostar MB-8425UUC-A Ver:3.1
9541	BTA	131510	AF	9536	AYS	N428S1	Shuttle HOT-433
9524	BTT	128410	AF	9536	AYS	N428C6	Shuttle HOT-433 (REV. 1)
9636	EYT	R80614	BF	9636	FXA	KM5Y54	? Biostar MB-8425UUD-A
9633	EYT	R62706	BF	9631	FXO	KM5Y52	Biostar MB-8425UUD-A
???	???	???445	???	???	???	???	QDI MP4-P4U885G
9623	EYS	R51132	BF	9625	FXT	KM5368	Biostar MB-8433UUD-A rev 3
9611	ETS	133940	BF	9615	FXS	RM3151	Biostar MB-8433UUD-A rev 2
9504	BTS	E12100	AF	9534	AYA	N522C6	PCChips M921
9519	BCS	N30514	AF	9520	AYT	N413G1	PCPartner UMC Lite 486
9540	ETA	135500	??stick	?	?	?	Sowah Research SR-M401-A
9523	BCT	N42646	F	9523	EYS	N42552	ECS UM8810P-AIO (rev 2.x)
9520	BCS	R40837	F	9442	AS	R90514	Chicony CH-881A REV 1.0
9521	BTA	124910	AF	9520	AYO	N33041	PCChips/Hsin Tech M915i
9524	BCO	N42643	AF	9540	BYS	R604B0	PCChips/Hsin Tech M921 (486)
9523	BCS	N42849	AF	9524	AYA	N511C5	TriGem SanJose
9518	BCS	N30701	AF	9519	AYS	N403L6	Shuttle HOT-433
???	BT?	???	A?	???	AY?	???	PCChips/Hsin Tech M915i, sanded markings (Bits und Bolts)
9514	BCA	N22726	AF	9521	AYA	N415B6	Shuttle HOT-433 (REV. 1)
9525	BTT	129700	AF	9520	AYO	N412J5	ADI F4DXP-UC5D-I

Suffix and revision analysis

Some chips may have sanded markings and they may be simply labeled as: “1/BT” and “6A/AY”. It is likely an equivalent of UM8881F-BTx and UM8886AF-AYx respectively, although the third letter of revision is uncertain (usually BTO, BTS and AYA, AYS, AYT respectively).

The omission of the third letter could point to a hypothesis the last letter doesn’t have any meaningful value on the board design (no HW nor SW differences). Or it could just lead to a laziness :-D.

On the other hand, the first weeks of manufacture of the HB have genuinely two letter revision codes, so it could be amount of manufactured chips (ex. more assembly lines).

The oldest HB seems to be from 9429 (BS or BA), after 9444 the 3-letter revision code appears (BCx, BTx). There was no Cxx code and only like two occurrences of Dxx code. The first occurrence of Exx code is from 9525 (Bxx revision was going in parallel). No Fxx ever appeared, so the development probably ended up with Exx revision. The main production of the HB probably ended in the last week of 1996 (last found datecode 9651). But during the exhaustive search a single datecode 9714 for revision EYS was found on PC-Chips board. It could be a custom batch or just selling accumulated parts. Interestingly there are revisions which are later in the alphabet.

The IBC has different part number suffixes (F, AF, BF and N). These have both SW and HW differences.

The oldest IBC which was found is F (suffix) from 9432. The first revisions were also two letters (AA,

AS). Since 9445 there are occurrences of F suffix with BS revision. The first AF suffix occurred in week 9450 with revision AYS. The F suffix is then manufactured in parallel with a revision EYS. The AF continues with revisions AYA, BYA and FYx until 9544. The first BF suffix appeared in 9539 with DYS revision. This was developed into FXS revision in 9611. Until the end of the general manufacturing in 9652 the revisions CYx and other FXx have appeared. The last suffix F occurred in 9638 with EYT revision.

No IBC with suffix N was stored in the database as these seems to be combined mostly with pentium generation HB.

A special case of 1997 date code was found thanks to Bits und Bolts in "Socket 3 Nightmare: New Old Stock, But Then I Bricked It!" video, there are 5x IBC chips of the latest date code even found. The markings are "UM8886BF 9724-FXT MM5G17". The serial codes were the same.

Serial code analysis

The "serial" code (the last line of the chip marking) is almost always different. However there were few chips with the same value.

Two HBs with the same "serial" code have other values same, with exception of "MA0508" (EYS vs EYA, these could be mistake from photos). The IBC follows the same rule.

There are chips with the same datecode, suffix and revision, which have nearly the same serial code. On the other hand, most of chips with the similar datecode have the serial code with high differences. There is a correlation between first 3 letters and datecode.

Taking all this into the account, the "serial" code may be a different way to encode suffix, revision, date, production line a/or a batch.

Existing PCI devices

Following PCI devices exist, according to the linux database:

VID	Vendor name
1060	United Microelectronics [UMC]

DID	Device name
0001	UM82C881
0002	UM82C886
0881	UM8881
0886	UM8886F
1001	UM886A
673a	UM8886BF
886a	UM8886A
8881	UM8881F
8886	UM8886F
888a	UM8886A
e881	UM8881N
e886	UM8886N
e88a	UM8886N

Chipset address ranges

Memory map:

Start	End	Usage
000c0000	000cffff	Segment C, ROM/ISA/RAM shadow, cacheable/writeable
000d0000	000dffff	Segment D, ROM/ISA/RAM shadow, cacheable/writeable
000e0000	000effff	Segment E, ROM/ISA/RAM shadow, cacheable/writeable
000f0000	000fffff	Segment F, ROM/ISA/RAM shadow, cacheable/writeable
fffc0000	fffcffff	Segment C, ROM/ISA, read only
fffd0000	fffdffff	Segment D, ROM/ISA, read only
fffe0000	fffeffff	Segment E, ROM/ISA, read only
ffff0000	ffffffff	Segment F, ROM/ISA, read only

Host Bridge (HB) UM8881-F

The Host Bridge is a PCI based 486 chipset. It integrates:

- 486 front side bus
- DRAM controller
- Up to L2 cache 1 MiB (dual bank/interleaved) or 512 kiB (single bank), write-back or write-through
- Tag RAM
- PCI busmaster
- VESA Local bus support
- Power management functions

No IDE controller is implemented in the HB. All communication with IBC and rest of the system is done via PCI bus.

HB revisions

The HB comes in multiple revisions. EDO seems to be supported only from PCI revision 4 and higher. Manual for Pine Technology PT-432B refers to the EDO capable HB as UM8881F/E, manual for Shuttle HOT-433 as UMC8881F-Exx.

The PCI revision can be also check via software. The 32-bit register `0x8` in PCI configuration space contains `Revision ID` field which shows the revision value.

NOTICE No HBs with values other than 4 were tested. This datasheet refers to HB PCI revision 4 in all documentation.

You can see the value if you execute pciutils's `lspci`:

```
00:10.0 Host bridge [0600]: United Microelectronics [UMC] UM8881F [1060:8881] (rev 04)
```

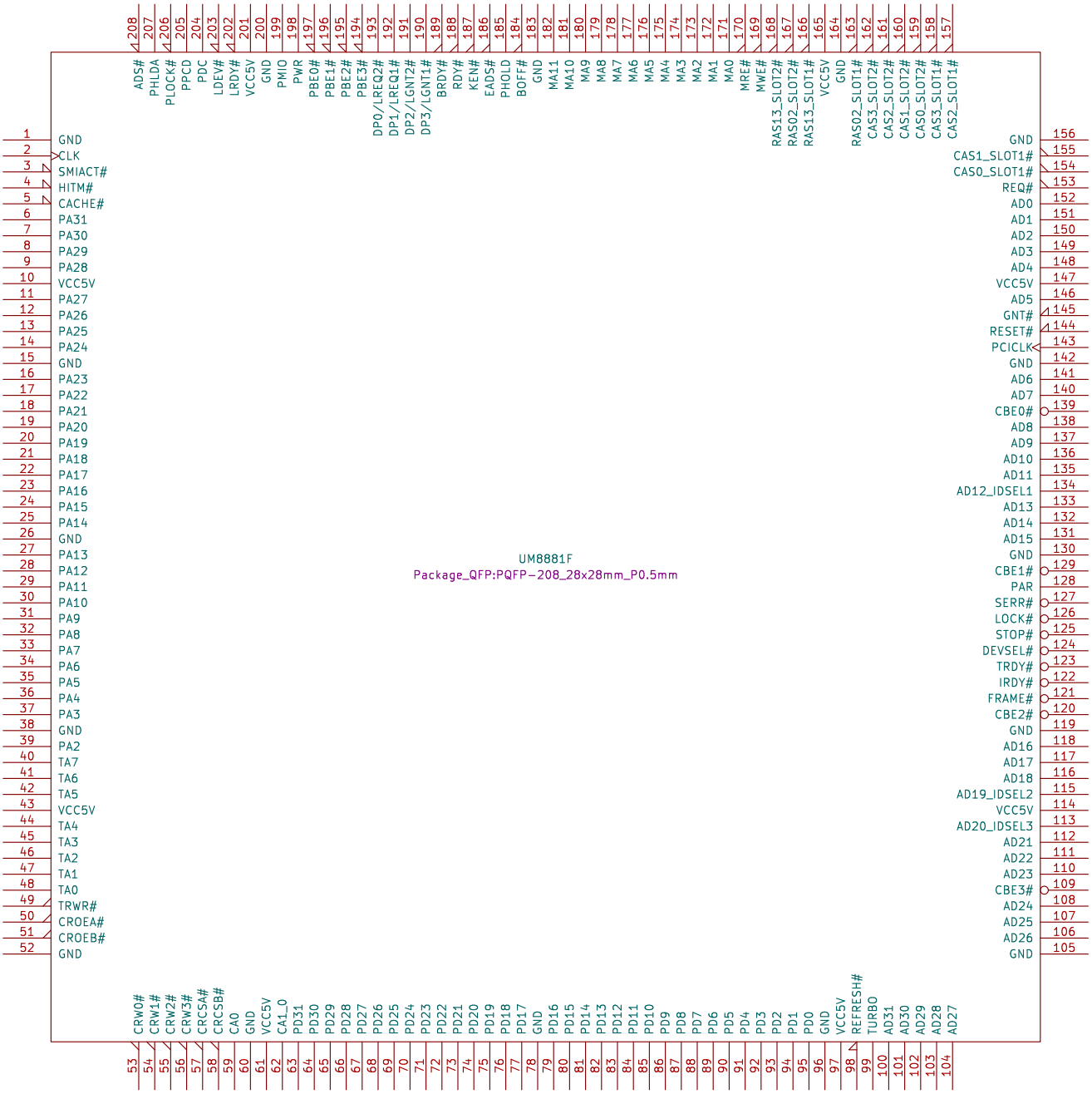
Or if you execute `PCICFG.EXE` in MSDOS:

```
PCI bus 00 device 10 function 00:  Header Type 'non-bridge' (single-func)
Vendor: 1060    United Microelectronics (UMC)
Device: 8881    UM8881 HB4 486 PCI Chipset
Class:    06    bridge                  Revision:   04
```

Other revisions may or may not have other HB registers readable/writable.

HB pin assignment

HB schematic symbol



HB pin list overview

1	GND
2	CLK
3	SMIACT#
4	HITM#
5	CACHE#
6	PA31
7	PA30
8	PA29
9	PA28
10	VCC5V
11	PA27
12	PA26
13	PA25
14	PA24
15	GND
16	PA23
17	PA22
18	PA21
19	PA20
20	PA19
21	PA18
22	PA17
23	PA16
24	PA15
25	PA14
26	GND
27	PA13
28	PA12
29	PA11
30	PA10
31	PA9
32	PA8
33	PA7
34	PA6
35	PA5
36	PA4
37	PA3
38	GND
39	PA2
40	TA7
41	TA6
42	TA5
43	VCC5V
44	TA4
45	TA3
46	TA2
47	TA1
48	TA0
49	TRWR#
50	CROEA#
51	CROEB#
52	GND

53	CRW0#
54	CRW1#
55	CRW2#
56	CRW3#
57	CRCSA#
58	CRCSB#
59	CA0
60	GND
61	VCC5V
62	CA1_0
63	PD31
64	PD30
65	PD29
66	PD28
67	PD27
68	PD26
69	PD25
70	PD24
71	PD23
72	PD22
73	PD21
74	PD20
75	PD19
76	PD18
77	PD17
78	GND
79	PD16
80	PD15
81	PD14
82	PD13
83	PD12
84	PD11
85	PD10
86	PD9
87	PD8
88	PD7
89	PD6
90	PD5
91	PD4
92	PD3
93	PD2
94	PD1
95	PD0
96	GND
97	VCC5V
98	REFRESH#
99	TURBO
100	AD31
101	AD30
102	AD29
103	AD28
104	AD27

105	GND
106	AD26
107	AD25
108	AD24
109	CBE3#
110	AD23
111	AD22
112	AD21
113	AD20_IDSEL3
114	VCC5V
115	AD19_IDSEL2
116	AD18
117	AD17
118	AD16
119	GND
120	CBE2#
121	FRAME#
122	IRDY#
123	TRDY#
124	DEVSEL#
125	STOP#
126	LOCK#
127	SERR#
128	PAR
129	CBE1#
130	GND
131	AD15
132	AD14
133	AD13
134	AD12_IDSEL1
135	AD11
136	AD10
137	AD9
138	AD8
139	CBE0#
140	AD7
141	AD6
142	GND
143	PCICLK
144	RESET#
145	GNT#
146	AD5
147	VCC5V
148	AD4
149	AD3
150	AD2
151	AD1
152	AD0
153	REQ#
154	CAS0_SLOT1#
155	CAS1_SLOT1#
156	GND

157	CAS2_SLOT1#
158	CAS3_SLOT1#
159	CAS0_SLOT2#
160	CAS1_SLOT2#
161	CAS2_SLOT2#
162	CAS3_SLOT2#
163	RAS02_SLOT1#
164	GND
165	VCC5V
166	RAS13_SLOT1#
167	RAS02_SLOT2#
168	RAS13_SLOT2#
169	MWE#
170	MRE#
171	MA0
172	MA1
173	MA2
174	MA3
175	MA4
176	MA5
177	MA6
178	MA7
179	MA8
180	MA9
181	MA10
182	MA11
183	GND
184	BOFF#
185	PHOLD
186	EADS#
187	KEN#
188	RDY#
189	BRDY#
190	DP3/LGNT1#
191	DP2/LGNT2#
192	DP1/LREQ1#
193	DP0/LREQ2#
194	PBE3#
195	PBE2#
196	PBE1#
197	PBE0#
198	PWR
199	PMIO
200	GND
201	VCC5V
202	LRDY#
203	LDEV#
204	PDC
205	PPCD
206	PLOCK#
207	PHLDA
208	ADS#

HB pin description

Pins were traced on A-Trend ATC-1415 board, so part's references are for this board.

Pin	Name	Direction	Type	Description
1	GND	power_in	line	Ground
2	CLK	input	clock	Chipset clock, via 10R + buffer to UM8886 pin 51
3	SMIACT#	input	input_low	Jumper J27, (also near J21 BREQ)
4	HITM#	input	input_low	multiple jumpers
5	CACHE#	input	input_low	J12.1 (via J12.2 to CPU), connected to J13.2 (via J13.1 to J26.2)
6	PA31	tri_state	line	CPU address 31
7	PA30	tri_state	line	CPU address 30
8	PA29	tri_state	line	CPU address 29
9	PA28	tri_state	line	CPU address 28
10	VCC5V	power_in	line	Power supply +5V
11	PA27	tri_state	line	CPU address 27
12	PA26	tri_state	line	CPU address 26
13	PA25	tri_state	line	CPU address 25
14	PA24	tri_state	line	CPU address 24
15	GND	power_in	line	Ground
16	PA23	tri_state	line	CPU address 23
17	PA22	tri_state	line	CPU address 22
18	PA21	tri_state	line	CPU address 21
19	PA20	tri_state	line	CPU address 20
20	PA19	tri_state	line	CPU/cache/tag address 19/17/15? TODO does 1MiB interleaved cache use pin for TAG address?
21	PA18	tri_state	line	CPU/cache/tag address 18/16/14, cache pin A16, tag pin A13/CE2 (via 512k 1-2 J1 jumper - closer to U1)
22	PA17	tri_state	line	CPU/cache/tag address 17/15/13, cache pin A15, tag pin A14/NC (via 256k 3-4 J1 jumper)
23	PA16	tri_state	line	CPU/cache/tag address 16/14/12, cache pin A14, tag pin A11
24	PA15	tri_state	line	CPU/cache/tag address 15/13/11, cache pin A13, tag pin A10
25	PA14	tri_state	line	CPU/cache/tag address 14/12/10, cache pin A2, tag pin A2
26	GND	power_in	line	Ground
27	PA13	tri_state	line	CPU/cache/tag address 13/11/9, cache pin A3, tag pin A3
28	PA12	tri_state	line	CPU/cache/tag address 12/10/8, cache pin A4, tag pin A4
29	PA11	tri_state	line	CPU/cache/tag address 11/9/7, cache pin A5, tag pin A5
30	PA10	tri_state	line	CPU/cache/tag address 10/8/6, cache pin A6, tag pin A6
31	PA9	tri_state	line	CPU/cache/tag address 9/7/5, cache pin A7, tag pin A7
32	PA8	tri_state	line	CPU/cache/tag address 8/6/4, cache pin A12, tag pin A12
33	PA7	tri_state	line	CPU/cache/tag address 7/5/3, cache pin A0, tag pin A0

Pin	Name	Direction	Type	Description
34	PA6	tri_state	line	CPU/cache/tag address 6/4/2, cache pin A1, tag pin A1
35	PA5	tri_state	line	CPU/cache/tag address 5/3/1, cache pin A8, tag pin A8
36	PA4	tri_state	line	CPU/cache/tag address 4/2/0, cache pin A9, tag pin A9
37	PA3	tri_state	line	CPU address 3
38	GND	power_in	line	Ground
39	PA2	tri_state	line	CPU address 2
40	TA7	bidirectional	line	Tag address, to tag data pin 7 (permutated?)
41	TA6	bidirectional	line	Tag address, to tag data pin 6 (permutated?)
42	TA5	bidirectional	line	Tag address, to tag data pin 5 (permutated?)
43	VCC5V	power_in	line	Power supply +5V
44	TA4	bidirectional	line	Tag address, to tag data pin 4 (permutated?)
45	TA3	bidirectional	line	Tag address, to tag data pin 3 (permutated?)
46	TA2	bidirectional	line	Tag address, to tag data pin 2 (permutated?)
47	TA1	bidirectional	line	Tag address, to tag data pin 1 (permutated?), bootstrap J5.2 via 2k2 to GND or VCC5V
48	TA0	bidirectional	line	Tag address, to tag data pin 0 (permutated?), bootstrap J4.2 via 2k2 to GND or VCC5V
49	TRWR#	output	output_low	Cache tag write enable
50	CROEA#	output	output_low	Cache output enable for bank A (or a single bank), via 33R
51	CROEB#	output	output_low	Cache output enable for bank B, possibly also via 33R (reported by mkarcher)
52	GND	power_in	line	Ground
53	CRW0#	output	output_low	Cache write enable [7:0], to U5 SRAM
54	CRW1#	output	output_low	Cache write enable [15:7], to U4 SRAM
55	CRW2#	output	output_low	Cache write enable [23:16], to U3 SRAM
56	CRW3#	output	output_low	Cache write enable [32:24], to U2 SRAM
57	CRCSA#	output	output_low	Cache chip select for bank A (or a single bank), via 33R
58	CRCSB#	output	output_low	Cache chip select for bank B, possibly also via 33R (reported by mkarcher)
59	CA0	output	line	Cache address 0 (single or first interleaved), pin A11
60	GND	power_in	line	Ground
61	VCC5V	power_in	line	Power supply +5V
62	CA1_0	output	line	Cache address 1 (single bank), address 0 (second interleaved), pin A10
63	PD31	tri_state	line	CPU (host) data 31
64	PD30	tri_state	line	CPU data 30
65	PD29	tri_state	line	CPU data 29
66	PD28	tri_state	line	CPU data 28
67	PD27	tri_state	line	CPU data 27
68	PD26	tri_state	line	CPU data 26
69	PD25	tri_state	line	CPU data 25
70	PD24	tri_state	line	CPU data 24
71	PD23	tri_state	line	CPU data 23
72	PD22	tri_state	line	CPU data 22
73	PD21	tri_state	line	CPU data 21
74	PD20	tri_state	line	CPU data 20
75	PD19	tri_state	line	CPU data 19
76	PD18	tri_state	line	CPU data 18
77	PD17	tri_state	line	CPU data 17

Pin	Name	Direction	Type	Description
78	GND	power_in	line	Ground
79	PD16	tri_state	line	CPU data 16
80	PD15	tri_state	line	CPU data 15
81	PD14	tri_state	line	CPU data 14
82	PD13	tri_state	line	CPU data 13
83	PD12	tri_state	line	CPU data 12
84	PD11	tri_state	line	CPU data 11
85	PD10	tri_state	line	CPU data 10
86	PD9	tri_state	line	CPU data 9
87	PD8	tri_state	line	CPU data 8
88	PD7	tri_state	line	CPU data 7
89	PD6	tri_state	line	CPU data 6
90	PD5	tri_state	line	CPU data 5
91	PD4	tri_state	line	CPU data 4
92	PD3	tri_state	line	CPU data 3
93	PD2	tri_state	line	CPU data 2
94	PD1	tri_state	line	CPU data 1
95	PD0	tri_state	line	CPU data 0
96	GND	power_in	line	Ground
97	VCC5V	power_in	line	Power supply +5V
98	REFRESH#	input	input_low	DRAM refresh, shared with ISA /REFRESH, via R22 (470R) to +5V, via R23 (33R) to UM8886 p145, via C7 to GND
99	TURBO	output	line	TODO, via R38 (10k) to +5V, to U18.2 (AND.I1), AND.I2 to UM8886 pin 54 (SMI2/LB2/KBCI), AND.Q via R88 to turbo LED
100	AD31	tri_state	line	PCI Address/Data 31
101	AD30	tri_state	line	PCI Address/Data 30
102	AD29	tri_state	line	PCI Address/Data 29
103	AD28	tri_state	line	PCI Address/Data 28
104	AD27	tri_state	line	PCI Address/Data 27
105	GND	power_in	line	Ground
106	AD26	tri_state	line	PCI Address/Data 26
107	AD25	tri_state	line	PCI Address/Data 25
108	AD24	tri_state	line	PCI Address/Data 24
109	CBE3#	tri_state	inverted	PCI command/byte enable 3
110	AD23	tri_state	line	PCI Address/Data 23
111	AD22	tri_state	line	PCI Address/Data 22
112	AD21	tri_state	line	PCI Address/Data 21
113	AD20_IDSEL3	tri_state	line	PCI Address/Data 20, IDSEL slot 3
114	VCC5V	power_in	line	Power supply +5V
115	AD19_IDSEL2	tri_state	line	PCI Address/Data 19, IDSEL slot 2
116	AD18	tri_state	line	PCI Address/Data 18
117	AD17	tri_state	line	PCI Address/Data 17
118	AD16	tri_state	line	PCI Address/Data 16
119	GND	power_in	line	Ground
120	CBE2#	tri_state	inverted	PCI command/byte enable 2
121	FRAME#	tri_state	inverted	PCI frame signal
122	IRDY#	tri_state	inverted	PCI initiator ready, via R17 (33R)
123	TRDY#	tri_state	inverted	PCI target ready, via R18 (33R)
124	DEVSEL#	tri_state	inverted	PCI target selected
125	STOP#	tri_state	inverted	PCI target halt request
126	LOCK#	tri_state	inverted	PCI locked transaction
127	SERR#	open_collector	inverted	PCI system error
128	PAR	tri_state	line	PCI parity

Pin	Name	Direction	Type	Description
129	CBE1#	tri_state	inverted	PCI command/byte enable 1
130	GND	power_in	line	Ground
131	AD15	tri_state	line	PCI Address/Data 15
132	AD14	tri_state	line	PCI Address/Data 14
133	AD13	tri_state	line	PCI Address/Data 13
134	AD12_IDSEL1	tri_state	line	PCI Address/Data 12, IDSEL slot 1
135	AD11	tri_state	line	PCI Address/Data 11
136	AD10	tri_state	line	PCI Address/Data 10
137	AD9	tri_state	line	PCI Address/Data 9
138	AD8	tri_state	line	PCI Address/Data 8
139	CBE0#	tri_state	inverted	PCI command/byte enable 0
140	AD7	tri_state	line	PCI Address/Data 7
141	AD6	tri_state	line	PCI Address/Data 6
142	GND	power_in	line	Ground
143	PCICLK	input	clock	PCI clock input driven via 10R and buffer from UM8886 pin 53
144	RESET#	input	input_low	PCI bus + main reset, via inverter (U10.2) from ISA RESET and UM8886 pin 157
145	GNT#	input	input_low	CPU gets granted PCI bus, driven from UM8886 pin 121
146	AD5	tri_state	line	PCI Address/Data 5
147	VCC5V	power_in	line	Power supply +5V
148	AD4	tri_state	line	PCI Address/Data 4
149	AD3	tri_state	line	PCI Address/Data 3
150	AD2	tri_state	line	PCI Address/Data 2
151	AD1	tri_state	line	PCI Address/Data 1
152	AD0	tri_state	line	PCI Address/Data 0
153	REQ#	output	output_low	CPU requests PCI bus, sent to UM8886 pin 124
154	CAS0_SLOT1#	output	output_low	DRAM column address strobe 0, SIMM slot 1 (possible permutations)
155	CAS1_SLOT1#	output	output_low	DRAM column address strobe 1, SIMM slot 1 (possible permutations)
156	GND	power_in	line	Ground
157	CAS2_SLOT1#	output	output_low	DRAM column address strobe 2, SIMM slot 1 (possible permutations)
158	CAS3_SLOT1#	output	output_low	DRAM column address strobe 3, SIMM slot 1 (possible permutations)
159	CAS0_SLOT2#	output	output_low	DRAM column address strobe 0, SIMM slot 2 (possible permutations)
160	CAS1_SLOT2#	output	output_low	DRAM column address strobe 1, SIMM slot 2 (possible permutations)
161	CAS2_SLOT2#	output	output_low	DRAM column address strobe 2, SIMM slot 2 (possible permutations)
162	CAS3_SLOT2#	output	output_low	DRAM column address strobe 3, SIMM slot 2 (possible permutations)
163	RAS02_SLOT1#	output	output_low	DRAM row address strobe 0/2, via buffer to SIMM slot 1 (possible permutations)
164	GND	power_in	line	Ground
165	VCC5V	power_in	line	Power supply +5V
166	RAS13_SLOT1#	output	output_low	DRAM row address strobe 1/3, via buffer to SIMM slot 1 (possible permutations)
167	RAS02_SLOT2#	output	output_low	DRAM row address strobe 0/2, via buffer to SIMM slot 2 (possible permutations)

Pin	Name	Direction	Type	Description
168	RAS13_SLOT2#	output	output_low	DRAM row address strobe 1/3, via buffer to SIMM slot 2 (possible permutations)
169	MWE#	output	output_low	DRAM write enable, via buffer and 10R to all SIMM slots
170	MRE#	output	output_low	Via 10R to 4x F245 pin T/R, optional buffer for DRAM data, (dual bank cache MB-4DUPC), TODO more tests
171	MA0	output	line	DRAM column/row address 0, via buffer
172	MA1	output	line	DRAM column/row address 1, via buffer
173	MA2	output	line	DRAM column/row address 2, via buffer
174	MA3	output	line	DRAM column/row address 3, via buffer
175	MA4	output	line	DRAM column/row address 4, via buffer
176	MA5	output	line	DRAM column/row address 5, via buffer
177	MA6	output	line	DRAM column/row address 6, via buffer
178	MA7	output	line	DRAM column/row address 7, via buffer
179	MA8	output	line	DRAM column/row address 8, via buffer
180	MA9	output	line	DRAM column/row address 9, via buffer
181	MA10	output	line	DRAM column/row address 10, via buffer
182	MA11	output	line	DRAM column/row address 11, via 10R
183	GND	power_in	line	Ground
184	BOFF#	output	output_low	CPU backoff from bus
185	PHOLD	output	line	CPU bus hold, pin E15
186	EADS#	output	output_low	CPU external address strobe
187	KEN#	output	output_low	CPU cache enable
188	RDY#	output	output_low	CPU non-burst ready, pin F16
189	BRDY#	output	output_low	CPU burst ready, pin H15
190	DP3/LGNT1#	bidirectional	line	CPU data parity 3, VLB local grant 1 (output)
191	DP2/LGNT2#	bidirectional	line	CPU data parity 2, VLB local grant 2 (output)
192	DP1/LREQ1#	bidirectional	line	CPU data parity 1, VLB local request 1 (input)
193	DP0/LREQ2#	bidirectional	line	CPU data parity 0, VLB local request 2 (input)
194	PBE3#	input	input_low	CPU byte enable 3
195	PBE2#	input	input_low	CPU byte enable 2
196	PBE1#	input	input_low	CPU byte enable 1
197	PBE0#	input	input_low	CPU byte enable 0
198	PWR	input	line	CPU write/read bus cycle, goes to jumpers (J15 INV, J17)
199	PMIO	input	line	CPU memory/IO bus cycle
200	GND	power_in	line	Ground
201	VCC5V	power_in	line	Power supply +5V
202	LRDY#	input	input_low	VLB target ready, unimplemented with a pull up via 4k7 (RP12) to VCC5V
203	LDEV#	input	input_low	VLB device cycle, unimplemented with a pull up via 4k7 (R52) to VCC5V, (boards use 10k too)
204	PDC	input	line	CPU data/control bus cycle, pin M15
205	PPCD	input	line	Page cache disable, pin J17
206	PLOCK#	input	input_low	CPU bus lock, pin N15
207	PHLDA	input	line	CPU hold acknowledge, pin P15
208	ADS#	input	input_low	CPU address status

HB registers

Register values (almost) after the reset:

```
50: 00 00 01 00 | 00 40 ff 0f | ff 0f 00 00 | 00 0f 00 ff
```

```
60: 00 00 00
```

Register description:

```
40 to 4F
        Not implemented (each returns 0x00 after writing 0xff)

50.80   L2 cache enable
        not sure if really enable (redundancy with size=none)
        May have sequential behavior
50.40   L2 cache mode (WT/WB)
50.30   Cache read burst (3-2-2-2/3-1-1-1/2-2-2-2/2-1-1-1)
        0: 3-2-2-2
        1: 3-1-1-1
        2: 2-2-2-2
        3: 2-1-1-1
50.08   Two banks of cache
50.07   L2 cache size (none/64K/128K/256K/512K/1024K/resvd/resvd)
        6-7 could be 2MiB
        TODO how the size and bankiness relates
        L2 cache must be made coherent with RAM, code cannot be run from cacheable region
            safe region is most likely ffff0000
        L1 invd will trigger L2 cache

51.C0   Read wait state (3/2/1/0)
        0: 3 states
        1: 2 states
        2: 1 state
        3: 0 states
51.30   Write wait state (3/2/1/0)
        0: 3 states
        1: 2 states
        2: 1 state
        3: 0 states
51.08   UNKNOWN, resource lock enable
51.04   A0000-BFFFF(?) PCI write merge
        possible:
        0: vl bus
        1: pci
51.02   Set after memory test, cleared for L1WB
        possible L1 WB policy
        0: WB
        1: WT
51.01   Tag allocation (7Tag+1Dirty/8Tag+0Dirty)

52.80   CPU to PCI Post Write
        0: 1 WS
        1: 0 WS
52.70   Upper region total capacity
        Is enabled only if lower region is enabled
        Set capacity must be always lower or the same as lower region or excess capacity will be reduced t
        Both banks in the region must have the same original capacity (or one bank with reduced or one ban
        If one bank is disabled, it must be re-sorted from larger to smaller (0)
        Upper bank could theoretically be configured as non-matching and useable if the non-implemented pa

        0: region disabled
        1/2/3/4/5/6/7 same as 52.07
```

```
52.08   Assign banks to lower and upper region
        0: lower region will define capacity for banks 0 and 1
            upper region will define capacity for banks 2 and 3
        1: lower region will define capacity for banks 2 and 3
            upper region will define capacity for banks 0 and 1
52.07   Lower region total capacity, physical memory address [0] to [L]
        sets capacity for a pair of banks (= SIMM slots)
        lower region _must_ have a larger capacity than upper region (use 52.08 if banks 2/3 are larger) -
        Lower region will always limit upper region to this capacity

        0: Both regions disabled (entire RAM inaccessible)
        1: 2 MiB
            ?? most likely 1x 9 cols, 10 rows, 61.30 = 00 (ignored)
        2: 4 MiB, 1M addresses
            1x 10 cols, 10 rows, 61.30 = 00
        3: 8 MiB, 2M addresses
            1x 10 cols, 11 rows, 61.30 = 00
            2x 10 cols, 10 rows, 61.30 = 00
        4: 16 MiB, 4M addresses
            1x 11 cols, 11 rows, 61.30 = 00
            2x 10 cols, 11 rows, 61.30 = 00
            1x 10 cols, 12 rows, 61.30 = 30
        5: 32 MiB
            2x 11 cols, 11 rows, 61.30 = 00
            2x 10 cols, 12 rows, 61.30 = 30
        6: 64 MiB
            NOTICE most likely too: 1x 12 cols, 12 rows, asymmetric
            4x 10 cols, 12 rows, 61.30 = 30
            4x 11 cols, 11 rows, 61.30 = 00
        7: 128 MiB
            NOTICE 4x 12 cols, 12 cols would make 256MiB for pair

        rest DRAMs untested (no samples)
        TODO how does older chipset revisions works here (probably no asymmetric support)

        (incomplete) extrapolated map [address -> pin] 1x single sided bank
        2-20 static for any mode (exact mapping unknown, likely linear)
        non 4k/symmetric
            next address bit is at new col or row as capacity increases
        4k/12 rows
            next address bits first in "rows" and then in "cols"

        pin     col     row
        --------------------
        MA0     2       11
        MA1     3       12
        MA2     4       13
        MA3     5       14
        MA4     6       15
        MA5     7       16
        MA6     8       17
        MA7     9       18
        MA8     10      19
        MA9     A       20
        MA10    C       B
        MA11    E       D
```

```
        RxC[as] A   B   C   D   E
        10x10   21  NC  NC  NC  NC
        10x11   21  22  NC  NC  NC
        11x11   21  22  23  NC  NC
        10x12as 23  21  24  NC  NC
        12x12as 23  21  24  22  25

        === 1x 32MiB: 10 cols, 12 rows, 2 sides, EDO ===
        52.07 = 05
        61.30 = 30
        === 1x 16MiB: 11 cols, 11 rows, 1 side, FPM ===
        52.07 = 04
        61.30 = 00
        === 1x 16MiB: 11 cols, 11 rows, 1 side, EDO ===
        52.07 = 04
        61.30 = 00
        === 1x 8MiB: 10 cols, 11 rows, 1 side, EDO ===
        52.07 = 03
        53.02 = 00
        61.30 = 00
        === 2x 8MiB: 10 cols, 11 rows, 1 side, EDO ===
        52.07 = 04
        53.02 = 01
        61.30 = 00

53.80   CPU to PCI Burst Write
53.40   Burst copy back option
53.20   swap bank 2/3 (= only bank 3 is used)
53.10   swap bank 0/1 (= only bank 1 is used)
53.08   Upper region sides
        0: single sided
        1: double sided
53.04   Upper region number of banks
        0: 1 bank occupied
        1: 2 banks occupied
53.02   Lower region sides
        0: single sided
        1: double sided
53.01   Lower region number of banks
        0: 1 bank occupied
        1: 2 banks occupied

54.80   DC000-DFFFF shadow read enable
54.40   D8000-DBFFF shadow read enable
54.20   D4000-D7FFF shadow read enable
54.10   D0000-D3FFF shadow read enable
54.08   CC000-CFFFF shadow read enable
54.04   C8000-CBFFF shadow read enable
54.02   C0000-C7FFF shadow read enable
54.01   E0000-EFFFF shadow read enable

55.80   Global shadow read enable (AND with 54 bits)
        F0000-FFFFF shadow read enable
        0: reads from ROM, writes to RAM (unless readonly)
        1: reads from RAM, writes to RAM (unless readonly)
```

```
        L2 cache should be initialized before
55.40   Global shadow write protect
        0: RAM will be always overwritten even if not shadowed
55.20   System BIOS cacheable
        only if shadow is enabled and write protection disabled
55.10   UNKNOWN, cleared in boot block (probably before flash)
55.08   UNKNOWN, cleared in boot block (probably before flash)
55.04   UNKNOWN, cleared in boot block (probably before flash)
55.02   UNKNOWN, cleared in boot block (probably before flash)
55.01   Video BIOS cacheable (TODO which addresses)

56.FF   Memory hole base (in 64K blocks)

57.80   Memory hole enabled/disabled
57.70   Memory hole size (64k/128k/256k/1M/2M/4M/8M/off)
        Can be used to "disable" caching (hole 0-2MiB)
57.0F   UNKNOWN

58.FF   0F if system BIOS cacheable

59.FF   00 if system BIOS cacheable
59.40   freezes linux if set

5A.80   UNKNOWN, crashes kernel
5A.40   Force L2 cache hit
        even if L2 is disabled (50.80)
        but not if RAM is disabled (probably needs computed cacheable range)
5A.20   UNKNOWN, setting seems to not affect L2
5A.10   Enable memory parity
5A.08   UNKNOWN, set to 1 if L2 cache disabled in BIOS (no effect on OS)
5A.04   UNKNOWN, set to 1 if L2 cache disabled in BIOS (no effect on OS)
5A.02   Some kind of deturbo
        could be always cache miss?
        all levels of memory seems to be slowed
        only if .08/04/01 is set
5A.01   Cleared during processor clock measurement

5B.FF   UNKNOWN, returns 0x2c on a booted linux, cannot be changed, but could be status

5C.FF   SMRAM base A27..A20

5D.80   Early Cache Write mode
5D.40   UNKNOWN
5D.20   UNKNOWN
5D.10   Slow Refresh
5D.0F   SMRAM base A31..A28, set to 0 in linux
        TODO enable PM in BIOS

5E.FF   UNKNOWN, writeable, set to 0x00 in linux

5F.FF   UNKNOWN, writeable, set to 0xff in linux

60.80   UNKNOWN
60.40   UNKNOWN
60.20   Set when enabling classic AMD SMRAM
60.10   UNKNOWN
```

```
60.08   UNKNOWN
60.04   UNKNOWN
60.02   Disable memory(?) parity
60.01   Open SMRAM space

61.C0   EDO mode? (no/resvd/resvd/yes)
61.20   Upper region asymmetric DRAM addressing
        Support for 12 columns (MA11)
        TODO document all modes (no 12x12 DRAM to test)
61.10   Lower region asymmetric DRAM addressing
        Support for 12 columns (MA11)
61.08   EDO speed (4-2-2-2/3-1-1-1)
61.04   UNKNOWN
61.02   Cyrix L1WB mode
61.01   UNKNOWN

62.FC   not implemented (cannot be set)
62.02   Burst mode (interleaved/linear)
        setting on am5x86 cause a crash
62.01   Set on early boot for revision "E"

63 to FF
        Not implemented (return 0x00 after writing 0xff)
```

ISA Bridge Controller (IBC) UM8886-F/AF/BF/N

The ISA Bridge Controller integrates most of the system's glue logic and bridges PCI bus to ISA bus. It's features includes:

- PCI arbiter for HB and regular PCI slots
- ISA bus
- PS/2 keyboard controller, optionally PS/2 mouse
- Minor CPU control signals
- IDE controller (only in AF/BF/N variants)
- Interface to RTC chip
- Up to 512 kiB flashable ROM BIOS
- IBM PC compatible chipset (PIT, PIC, DMA, PPI) logic
- Power management logic
- RESET logic
- Clock distribution

IBC suffixes

The IBC comes in 4 confirmed different variants marked with F/AF/BF/N suffixes.

It is highly probable F vs AF/BF will differ in pinout as F doesn't implement any IDE controller. No same motherboard models with F/AF variability were found.

Parts suffixed by AF and BF seems to be almost fully compatible without a huge board redesign. Boards A-Trend ATC-1415, PCChips/Hsin Tech M921 (486) and PCPartner UMC Lite 486 seem to vary between both variants. There may be some invisible changes on PCB though.

BF vs N variant definitely changes some pins functionality including power supply pins. N variant seems to have crammed support for devices on notebooks (ex. encoded DMA requests, docking and PC card signals and differently located PS/2 controllers). On ATC-1415 board with BF variant the keyboard clock is connected to UM8886BF pin 57 but schematics for UM8886-N has this pin labeled as PGP1/LDEV#/GA20, the keyboard data on BF leads to pin 56, which is called PGP0(RC#*) on variant

N. UM8886N can have the keyboard input on pin 54, which is labeled “SMI2#/LB2#/KBCI”. Also as Pentium boards can use 66 MHz FSB, some parts of this variant will be clocked at this speed.

AF variant was found on a pentium board. It seems UM8886 is universal enough to be used as a generic PCI/ISA bridge.

IDE is implemented only in AF/BF/N suffixed chips. The boards with F chip often use CMD640 or other IDE controller solutions.

From the software point of view all variants differs. The AF has configuration registers for IDE located at I/O port `0x108`. The BF IDE controller uses a PCI config space. Also the BF variant can remap the pin usage for PS/2 mouse (shared with ISA bus address signals). The N variant was not tested (no HW). But from the schematics it is built as an evolution from BF variant. It also needs to have additional configuration registers for notebook specific designs.

IBC revisions

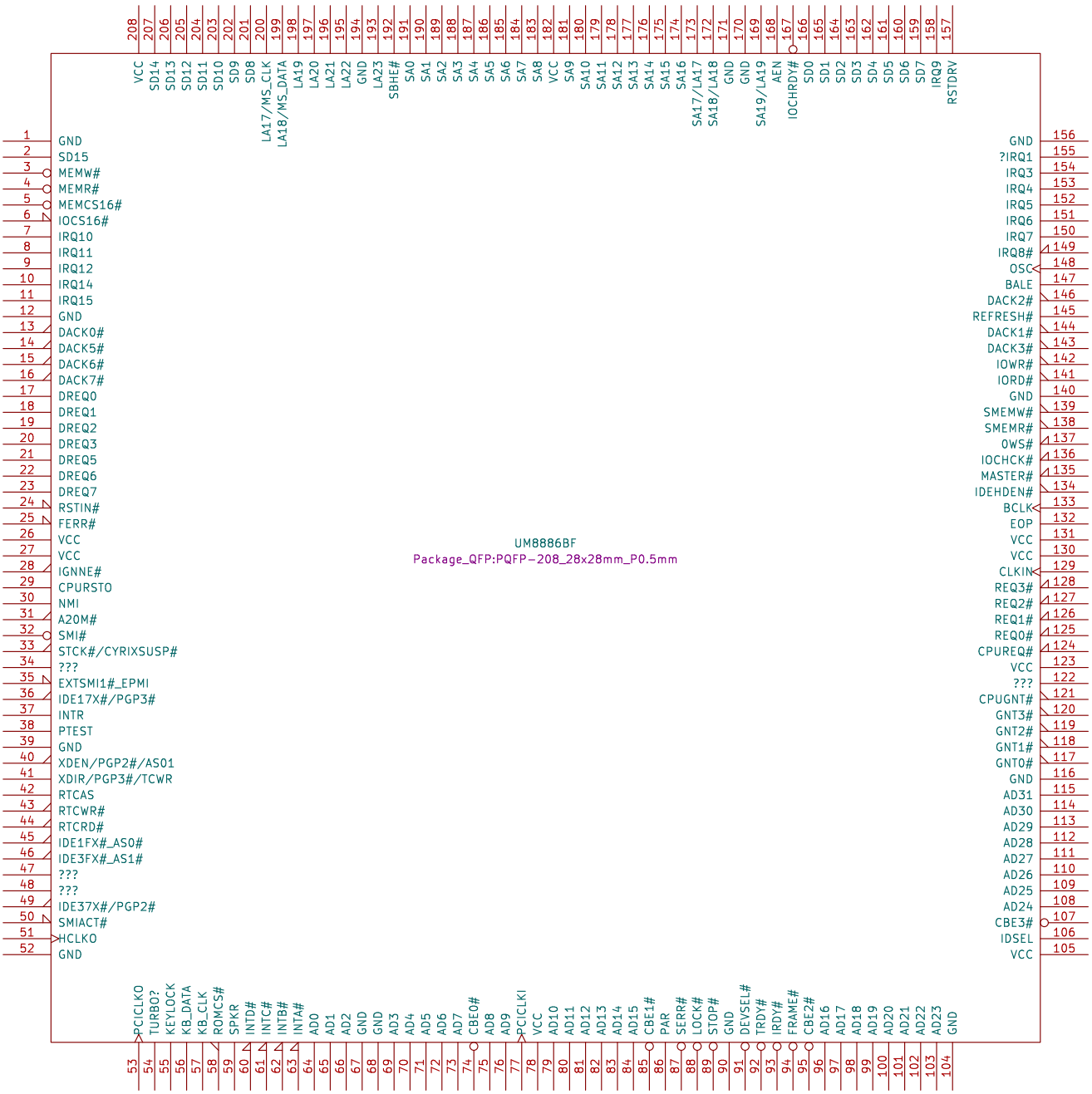
As for revisions, no exact problems were reported, but it is generally recommended to have as latest revision as possible :-).

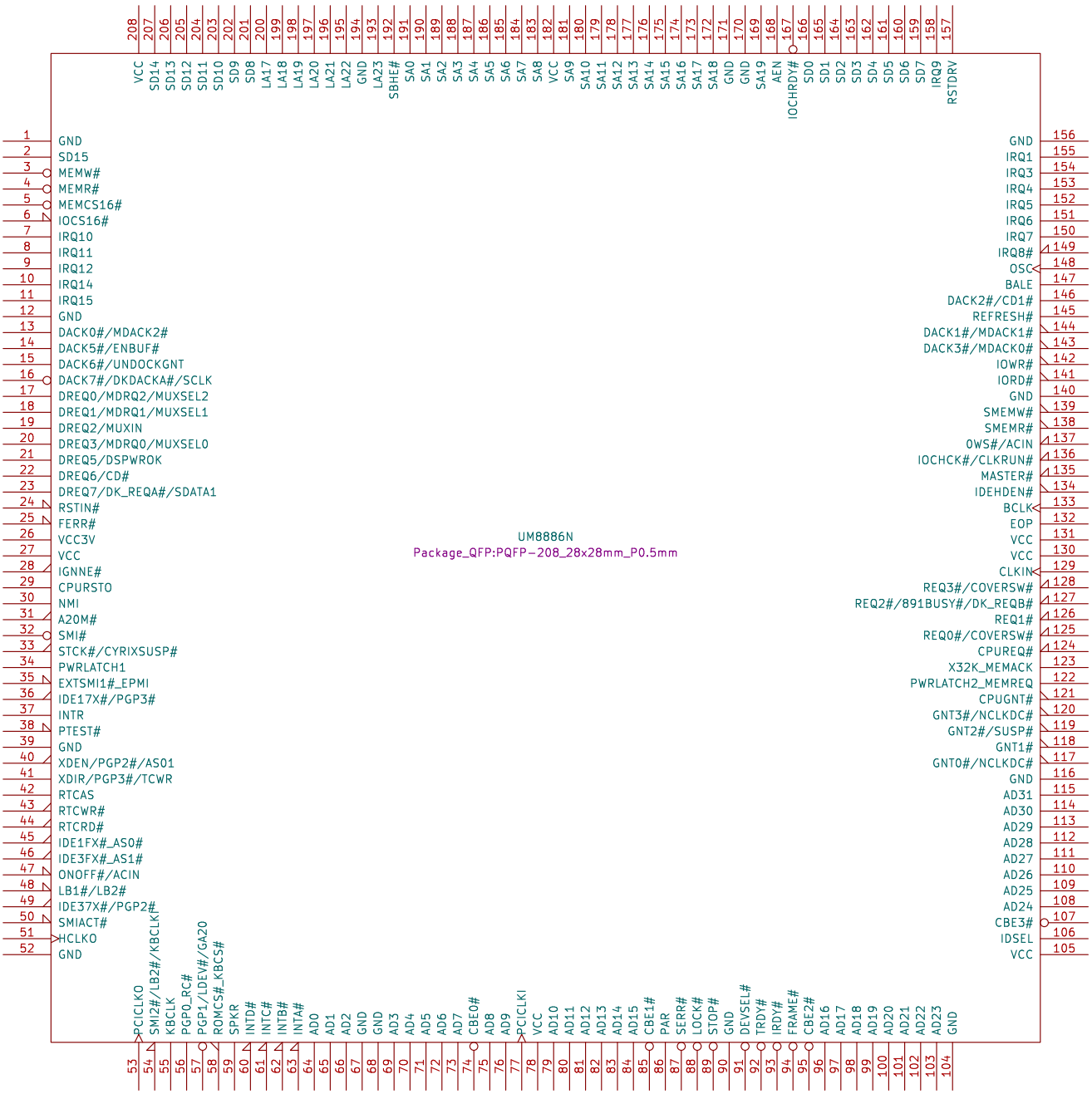
It is possible i8042 controller in `UM8886BF 9602 DYS` doesn’t support `0xd3` command (= write byte to the second PS/2 port output buffer). This will fail autodetection in linux kernel. It is possible BIOS fixes this by emulating the command in an SMM handler. The `UM8886BF 9602 DYS` PCI arbiter supports PCI bus masters behind PCI (and even PCIe) bridge cascade, but older parts may have bugs in its PCI implementation (other 486 chipsets introduce multiple bus deadlocks if used with PCI bridges). IDE support (especially higher speeds) may have bugs too.

IBC pin assignment

IBC schematic symbol

No F nor AF variants pinouts were traced. N variant pinout was researched from Mitac schematics.


UM8886BF
Package_QFP:PQFP-208_28x28mm_P0.5mm
1 GND
2 SD15
3 MEMW#
4 MEMR#
5 MEMCS16#
6 IOCS16#
7 IRQ10
8 IRQ11
9 IRQ12
10 IRQ14
11 IRQ15
12 GND
13 DACK0#
14 DACK5#
15 DACK6#
16 DACK7#
17 DREQ0
18 DREQ1
19 DREQ2
20 DREQ3
21 DREQ5
22 DREQ6
23 DREQ7
24 RSTIN#
25 FERR#
26 VCC
27 VCC
28 IGNNE#
29 CPURSTO
30 NMI
31 A20M#
32 SMI#
33 STCK#/CYRIXSUSP#
34 ???
35 EXTSMI1#_EPMI
36 IDE17X#/PGP3#
37 INTR
38 PTEST
39 GND
40 XDEN/PGP2#/AS01
41 XDIR/PGP3#/TCWR
42 RTCAS
43 RTCWR#
44 RTCRD#
45 IDE1FX#_AS0#
46 IDE3FX#_AS1#
47 ???
48 ???
49 IDE37X#/PGP2#
50 SMIACT#
51 HCLKO
52 GND
53 PCICLKO
54 TURBO?
55 KEYLOCK
56 KB_DATA
57 KB_CLK
58 ROMCS#
59 SPKR
60 INTD#
61 INTC#
62 INTB#
63 INTA#
64 AD0
65 AD1
66 AD2
67 GND
68 GND
69 AD3
70 AD4
71 AD5
72 AD6
73 AD7
74 CBE0#
75 AD8
76 AD9
77 PCICLKI
78 VCC
79 AD10
80 AD11
81 AD12
82 AD13
83 AD14
84 AD15
85 CBE1#
86 PAR
87 SERR#
88 LOCK#
89 STOP#
90 GND
91 DEVSEL#
92 TRDY#
93 IRDY#
94 FRAME#
95 CBE2#
96 AD16
97 AD17
98 AD18
99 AD19
100 AD20
101 AD21
102 AD22
103 AD23
104 GND
105 VCC
106 IDSEL
107 CBE3#
108 AD24
109 AD25
110 AD26
111 AD27
112 AD28
113 AD29
114 AD30
115 AD31
116 GND
117 GNT0#
118 GNT1#
119 GNT2#
120 GNT3#
121 CPUGNT#
122 ???
123 VCC
124 CPUREQ#
125 REQ0#
126 REQ1#
127 REQ2#
128 REQ3#
129 CLKIN
130 VCC
131 VCC
132 EOP
133 BCLK
134 IDEHDEN#
135 MASTER#
136 IOCHCK#
137 OWS#
138 SMEMR#
139 SMEMW#
140 GND
141 IORD#
142 IOWR#
143 DACK3#
144 DACK1#
145 REFRESH#
146 DACK2#
147 BALE
148 OSC
149 IRQ8#
150 IRQ7
151 IRQ6
152 IRQ5
153 IRQ4
154 IRQ3
155 ?IRQ1
156 GND
157 RSTDRV
158 IRQ9
159 SD7
160 SD6
161 SD5
162 SD4
163 SD3
164 SD2
165 SD1
166 SD0
167 IOCHRDY#
168 AEN
169 SA19/LA19
170 GND
171 GND
172 SA18/LA18
173 SA17/LA17
174 SA16
175 SA15
176 SA14
177 SA13
178 SA12
179 SA11
180 SA10
181 SA9
182 VCC
183 SA8
184 SA7
185 SA6
186 SA5
187 SA4
188 SA3
189 SA2
190 SA1
191 SA0
192 SBHE#
193 LA23
194 GND
195 LA22
196 LA21
197 LA20
198 LA19
199 LA18/MS_DATA
200 LA17/MS_CLK
201 SD8
202 SD9
203 SD10
204 SD11
205 SD12
206 SD13
207 SD14
208 VCC


UM8886N
Package_QFP:PQFP-208_28x28mm_P0.5mm
1 GND
2 SD15
3 MEMW#
4 MEMR#
5 MEMCS16#
6 IOCS16#
7 IRQ10
8 IRQ11
9 IRQ12
10 IRQ14
11 IRQ15
12 GND
13 DACK0#/MDACK2#
14 DACK5#/ENBUF#
15 DACK6#/UNDOCKGNT
16 DACK7#/DKDACKA#/SCLK
17 DREQ0/MDRQ2/MUXSEL2
18 DREQ1/MDRQ1/MUXSEL1
19 DREQ2/MUXIN
20 DREQ3/MDRQ0/MUXSEL0
21 DREQ5/DSPWROK
22 DREQ6/CD#
23 DREQ7/DK_REQA#/SDATA1
24 RSTIN#
25 FERR#
26 VCC3V
27 VCC
28 IGNNE#
29 CPURSTO
30 NMI
31 A20M#
32 SMI#
33 STCK#/CYRIXSUSP#
34 PWRLATCH1
35 EXTSMI1#_EPMI
36 IDE17X#/PGP3#
37 INTR
38 PTEST#
39 GND
40 XDEN/PGP2#/AS01
41 XDIR/PGP3#/TCWR
42 RTCAS
43 RTCWR#
44 RTCRD#
45 IDE1FX#_AS0#
46 IDE3FX#_AS1#
47 ONOFF#/ACIN
48 LB1#/LB2#
49 IDE37X#/PGP2#
50 SMIACT#
51 HCLKO
52 GND
53 PCICLKO
54 SMI2#/LB2#/KBCLKI
55 KBCLK
56 PGPO_RC#
57 PGP1/LDEV#/GA20
58 ROMCS#_KBCS#
59 SPKR
60 INTD#
61 INTC#
62 INTB#
63 INTA#
64 AD0
65 AD1
66 AD2
67 GND
68 GND
69 AD3
70 AD4
71 AD5
72 AD6
73 AD7
74 CBE0#
75 AD8
76 AD9
77 PCICLKI
78 VCC
79 AD10
80 AD11
81 AD12
82 AD13
83 AD14
84 AD15
85 CBE1#
86 PAR
87 SERR#
88 LOCK#
89 STOP#
90 GND
91 DEVSEL#
92 TRDY#
93 IRDY#
94 FRAME#
95 CBE2#
96 AD16
97 AD17
98 AD18
99 AD19
100 AD20
101 AD21
102 AD22
103 AD23
104 GND
105 VCC
106 IDSEL
107 CBE3#
108 AD24
109 AD25
110 AD26
111 AD27
112 AD28
113 AD29
114 AD30
115 AD31
116 GND
117 GNT0#/NCLKDC#
118 GNT1#
119 GNT2#/SUSP#
120 GNT3#/NCLKDC#
121 CPUGNT#
122 PWRLATCH2_MEMREQ
123 X32K_MEMACK
124 CPUREQ#
125 REQ0#/COVERSW#
126 REQ1#
127 REQ2#/891BUSY#/DK_REQB#
128 REQ3#/COVERSW#
129 CLKIN
130 VCC
131 VCC
132 EOP
133 BCLK
134 IDEHDEN#
135 MASTER#
136 IOCHCK#/CLKRUN#
137 OWS#/ACIN
138 SMEMR#
139 SMEMW#
140 GND
141 IORD#
142 IOWR#
143 DACK3#/MDACK0#
144 DACK1#/MDACK1#
145 REFRESH#
146 DACK2#/CD1#
147 BALE
148 OSC
149 IRQ8#
150 IRQ7
151 IRQ6
152 IRQ5
153 IRQ4
154 IRQ3
155 IRQ1
156 GND
157 RSTDRV
158 IRQ9
159 SD7
160 SD6
161 SD5
162 SD4
163 SD3
164 SD2
165 SD1
166 SD0
167 IOCHRDY#
168 AEN
169 SA19
170 GND
171 GND
172 SA18
173 SA17
174 SA16
175 SA15
176 SA14
177 SA13
178 SA12
179 SA11
180 SA10
181 SA9
182 VCC
183 SA8
184 SA7
185 SA6
186 SA5
187 SA4
188 SA3
189 SA2
190 SA1
191 SA0
192 SBHE#
193 LA23
194 GND
195 LA22
196 LA21
197 LA20
198 LA19
199 LA18
200 LA17
201 SD8
202 SD9
203 SD10
204 SD11
205 SD12
206 SD13
207 SD14
208 VCC

IBC UM8886-BF pin list overview

1	GND
2	SD15
3	MEMW#
4	MEMR#
5	MEMCS16#
6	IOCS16#
7	IRQ10
8	IRQ11
9	IRQ12
10	IRQ14
11	IRQ15
12	GND
13	DACK0#
14	DACK5#
15	DACK6#
16	DACK7#
17	DREQ0
18	DREQ1
19	DREQ2
20	DREQ3
21	DREQ5
22	DREQ6
23	DREQ7
24	RSTIN#
25	FERR#
26	VCC
27	VCC
28	IGNNE#
29	CPURSTO
30	NMI
31	A20M#
32	SMI#
33	STCK#/CYRIXSUSP#
34	???
35	EXTSMI1#_EPMI
36	IDE17X#/PGP3#
37	INTR
38	PTEST#
39	GND
40	XDEN/PGP2#/AS01
41	XDIR/PGP3#/TCWR
42	RTCAS
43	RTCWR#
44	RTCRD#
45	IDE1FX#_AS0#
46	IDE3FX#_AS1#
47	???
48	???
49	IDE37X#/PGP2#
50	SMIACT#
51	HCLKO
52	GND

53	PCICLKO
54	TURBO?
55	KEYLOCK
56	KB_DATA
57	KB_CLK
58	ROMCS#
59	SPKR
60	INTD#
61	INTC#
62	INTB#
63	INTA#
64	AD0
65	AD1
66	AD2
67	GND
68	GND
69	AD3
70	AD4
71	AD5
72	AD6
73	AD7
74	CBE0#
75	AD8
76	AD9
77	PCICLKI
78	VCC
79	AD10
80	AD11
81	AD12
82	AD13
83	AD14
84	AD15
85	CBE1#
86	PAR
87	SERR#
88	LOCK#
89	STOP#
90	GND
91	DEVSEL#
92	TRDY#
93	IRDY#
94	FRAME#
95	CBE2#
96	AD16
97	AD17
98	AD18
99	AD19
100	AD20
101	AD21
102	AD22
103	AD23
104	GND

105	VCC
106	IDSEL
107	CBE3#
108	AD24
109	AD25
110	AD26
111	AD27
112	AD28
113	AD29
114	AD30
115	AD31
116	GND
117	GNT0#
118	GNT1#
119	GNT2#
120	GNT3#
121	CPUGNT#
122	???
123	VCC
124	CPUREQ#
125	REQ0#
126	REQ1#
127	REQ2#
128	REQ3#
129	CLKIN
130	VCC
131	VCC
132	EOP
133	BCLK
134	IDEHDEN#
135	MASTER#
136	IOCHCK#
137	0WS#
138	SMEMR#
139	SMEMW#
140	GND
141	IORD#
142	IOWR#
143	DACK3#
144	DACK1#
145	REFRESH#
146	DACK2#
147	BALE
148	OSC
149	IRQ8#
150	IRQ7
151	IRQ6
152	IRQ5
153	IRQ4
154	IRQ3
155	?IRQ1
156	GND

157	RSTDRV
158	IRQ9
159	SD7
160	SD6
161	SD5
162	SD4
163	SD3
164	SD2
165	SD1
166	SD0
167	IOCHRDY#
168	AEN
169	SA19/LA19
170	GND
171	GND
172	SA18/LA18
173	SA17/LA17
174	SA16
175	SA15
176	SA14
177	SA13
178	SA12
179	SA11
180	SA10
181	SA9
182	VCC
183	SA8
184	SA7
185	SA6
186	SA5
187	SA4
188	SA3
189	SA2
190	SA1
191	SA0
192	SBHE#
193	LA23
194	GND
195	LA22
196	LA21
197	LA20
198	LA19
199	LA18/MS_DATA
200	LA17/MS_CLK
201	SD8
202	SD9
203	SD10
204	SD11
205	SD12
206	SD13
207	SD14
208	VCC

IBC UM8886-BF pin description

Pins were traced on A-Trend ATC-1415 board and partially on Biostar MB-8433UUD-A board. The references to circuit parts should be attributed to the respective board.

Data bus for BIOS and address/data bus for RTC on ATC-1415 are buffered via 74F245 and then they are connected to ISA data pins. BIOS/RTC cycle is distinguished from ISA cycle by DIR and /OE pins of the buffer. Buffer 74F245 direction pin 1 is driven from UM8886-BF pin 41 XDIR. 74F245's output enable pin 19 is driven from UM8886-BF pin 40 XDEN.

The IBC creates only control signals of the IDE interface, the data signals are shared with the ISA bus (via 74F245 buffers). The IDE bus cycle (primary or secondary) is distinguished from an ISA bus cycle with /OE and DIR pins of F245 buffers.

Pin	Name	Direction	Type	Description
1	GND	power_in	line	Ground
2	SD15	tri_state	line	ISA data 15
3	MEMW#	tri_state	inverted	ISA memory write (pin C10)
4	MEMR#	tri_state	inverted	ISA memory read (pin C9)
5	MEMCS16#	tri_state	inverted	ISA 16bit memory chip select
6	IOCS16#	input	input_low	ISA 16bit I/O chip select
7	IRQ10	input	line	ISA interrupt 10
8	IRQ11	input	line	ISA interrupt 11
9	IRQ12	input	line	ISA interrupt 12
10	IRQ14	input	line	ISA interrupt 14
11	IRQ15	input	line	ISA interrupt 15
12	GND	power_in	line	Ground
13	DACK0#	output	output_low	ISA DMA acknowledge 0
14	DACK5#	output	output_low	ISA DMA acknowledge 5
15	DACK6#	output	output_low	ISA DMA acknowledge 6
16	DACK7#	output	output_low	ISA DMA acknowledge 7
17	DREQ0	input	line	ISA DMA request 0
18	DREQ1	input	line	ISA DMA request 1
19	DREQ2	input	line	ISA DMA request 2
20	DREQ3	input	line	ISA DMA request 3
21	DREQ5	input	line	ISA DMA request 5
22	DREQ6	input	line	ISA DMA request 6
23	DREQ7	input	line	ISA DMA request 7
24	RSTIN#	input	input_low	?Reset input from power good circuitry, R89 on ATC-1415
25	FERR#	input	input_low	FPU error
26	VCC	power_in	line	Power supply +5V (possibly 3.3V optionally)
27	VCC	power_in	line	Power supply +5V
28	IGNNE#	output	output_low	Ignore FPU error
29	CPURSTO	output	line	CPU reset (486 pin C10 SRST)
30	NMI	output	line	Nonmaskable interrupt to CPU
31	A20M#	output	output_low	Mask CPU address 20
32	SMI#	bidirectional	inverted	System management interrupt, output request, input SMM status (486 pin B10 SMI)
33	STCK#/CYRIXSUSP#	output	output_low	CPU stop clock / ?Cyrix suspend (486 pin G15 STPCLK)
34	???	???		???
35	EXTSMI1#_EPMI	input	input_low	External SMI 1 (PMU switch on ATC-1415)
36	IDE17X#/PGP3#	output	output_low	IDE 17Xh chip select (secondary) / Programmable output pin 3
37	INTR	output	line	Maskable CPU interrupt (486 pin A16 INTR)

Pin	Name	Direction	Type	Description
38	PTEST#	???		via R89 to pin 24 (RSTIN), PTEST on revision N
39	GND	power_in	line	Ground
40	XDEN/PGP2#/AS01	output	output_low	XD bus (8bit ISA) data buffer enable (BIOS 74F245 nOE) / Programmable output pin 3 / ???
41	XDIR/PGP3#/TCWR	output	line	XD bus (8bit ISA) data buffer direction (BIOS 74F245 DIR) / Programmable output pin 2 / RTC 4K RAM R/W signal
42	RTCAS	output	line	RTC address latch
43	RTCWR#	output	output_low	RTC write (RTC chip pin 14 R/W on ATC-1415)
44	RTCRD#	output	output_low	RTC read (RTC chip pin 17 DS on ATC-1415)
45	IDE1FX#_AS0#	output	output_low	IDE 1FXh chip select (primary)
46	IDE3FX#_AS1#	output	output_low	IDE 3FXh chip select (primary)
47	???	?output		?? (ATC-1415 open jumper to RTC chip pin 18 nRESET)
48	???	???		???
49	IDE37X#/PGP2#	output	output_low	IDE 37Xh chip select (secondary) / Programmable output pin 2
50	SMIACT#	input	input_low	System management interrupt acknowledge from CPU (486 pin C12 SMIACT)
51	HCLKO	output	clock	Global clock output to CPU and other parts of the chipset (on ATC-1415 via F244 buffers)
52	GND	power_in	line	Ground
53	PCICLKO	output	clock	Global clock output to the PCI subsystem
54	TURBO?	open_collector	line	ATC-1415: turbo switch AND gate, pullup
55	KEYLOCK	input	line	Keylock switch
56	KB_DATA	tri_state	line	AT keyboard data (input and output seems to be integrated in a single pin)
57	KB_CLK	tri_state	line	AT keyboard clock (input and output seems to be integrated in a single pin)
58	ROMCS#	output	output_low	ROM chip select (keyboard controller chip select uncertain)
59	SPKR	output	line	Speaker output
60	INTD#	input	input_low	PCI interrupt D
61	INTC#	input	input_low	PCI interrupt C
62	INTB#	input	input_low	PCI interrupt B
63	INTA#	input	input_low	PCI interrupt A
64	AD0	tri_state	line	PCI Address/Data 0
65	AD1	tri_state	line	PCI Address/Data 1
66	AD2	tri_state	line	PCI Address/Data 2
67	GND	power_in	line	Ground
68	GND	power_in	line	Ground
69	AD3	tri_state	line	PCI Address/Data 3
70	AD4	tri_state	line	PCI Address/Data 4
71	AD5	tri_state	line	PCI Address/Data 5
72	AD6	tri_state	line	PCI Address/Data 6
73	AD7	tri_state	line	PCI Address/Data 7
74	CBE0#	tri_state	inverted	PCI command/byte enable 0
75	AD8	tri_state	line	PCI Address/Data 8
76	AD9	tri_state	line	PCI Address/Data 9
77	PCICLKI	input	clock	PCI clock input (positive edge)
78	VCC	power_in	line	Power supply +5V
79	AD10	tri_state	line	PCI Address/Data 10
80	AD11	tri_state	line	PCI Address/Data 11

Pin	Name	Direction	Type	Description
81	AD12	tri_state	line	PCI Address/Data 12
82	AD13	tri_state	line	PCI Address/Data 13
83	AD14	tri_state	line	PCI Address/Data 14
84	AD15	tri_state	line	PCI Address/Data 15
85	CBE1#	tri_state	inverted	PCI command/byte enable 1
86	PAR	tri_state	line	PCI parity bit
87	SERR#	input	inverted	PCI system error, will generate NMI
88	LOCK#	input	inverted	PCI lock signal
89	STOP#	tri_state	inverted	PCI stop signal
90	GND	power_in	line	Ground
91	DEVSEL#	tri_state	inverted	PCI device select
92	TRDY#	tri_state	inverted	PCI target ready
93	IRDY#	tri_state	inverted	PCI Initiator ready
94	FRAME#	tri_state	inverted	PCI frame signal
95	CBE2#	tri_state	inverted	PCI command/byte enable 2
96	AD16	tri_state	line	PCI Address/Data 16
97	AD17	tri_state	line	PCI Address/Data 17
98	AD18	tri_state	line	PCI Address/Data 18
99	AD19	tri_state	line	PCI Address/Data 19
100	AD20	tri_state	line	PCI Address/Data 20
101	AD21	tri_state	line	PCI Address/Data 21
102	AD22	tri_state	line	PCI Address/Data 22
103	AD23	tri_state	line	PCI Address/Data 23
104	GND	power_in	line	Ground
105	VCC	power_in	line	Power supply +5V
106	IDSEL	input	line	PCI ID select from northbridge, connects to AD18
107	CBE3#	tri_state	inverted	PCI command/byte enable 3
108	AD24	tri_state	line	PCI Address/Data 24
109	AD25	tri_state	line	PCI Address/Data 25
110	AD26	tri_state	line	PCI Address/Data 26
111	AD27	tri_state	line	PCI Address/Data 27
112	AD28	tri_state	line	PCI Address/Data 28
113	AD29	tri_state	line	PCI Address/Data 29
114	AD30	tri_state	line	PCI Address/Data 30
115	AD31	tri_state	line	PCI Address/Data 31
116	GND	power_in	line	Ground
117	GNT0#	output	output_low	PCI grant to master 0 (ATC-1415 PCI1 slot)
118	GNT1#	output	output_low	PCI grant to master 1 (ATC-1415 PCI2 slot)
119	GNT2#	output	output_low	PCI grant to master 2 (ATC-1415 PCI3 slot)
120	GNT3#	output	output_low	PCI grant to master 3 (ATC-1415 just Vcc pullup)
121	CPUGNT#	output	output_low	PCI CPU grant (connects to p145 NB)
122	???	???		???
123	VCC	power_in	line	Power supply +5V (seems to be valid Vcc, including caps)
124	CPUREQ#	input	input_low	PCI request from CPU (connects to p153 NB)
125	REQ0#	input	input_low	PCI request from master 0 (ATC-1415 PCI1 slot)
126	REQ1#	input	input_low	PCI request from master 1 (ATC-1415 PCI2 slot)
127	REQ2#	input	input_low	PCI request from master 2 (ATC-1415 PCI3 slot)
128	REQ3#	input	input_low	PCI request from master 3 (ATC-1415 just Vcc pullup)

Pin	Name	Direction	Type	Description
129	CLKIN	input	clock	Main posedge clock input from generator (VT8235N pin 8 on ATC-1415)
130	VCC	power_in	line	Power supply +5V
131	VCC	power_in	line	Power supply +5V
132	EOP	bidir	line	End of DMA process
133	BCLK	output	clock	ISA SYSCLK
134	IDEHDEN#	output	output_low	IDE cycle enable (for ISA/IDE buffer) (ATC-1415 /OE F245 buffer)
135	MASTER#	input	input_low	ISA 16bit master drives bus
136	IOCHCK#	input	input_low	ISA I/O channel check (will generate NMI)
137	0WS#	input	input_low	Zero wait states request from ISA
138	SMEMR#	output	output_low	ISA system memory read (B12)
139	SMEMW#	output	output_low	ISA system memory write (B11)
140	GND	power_in	line	Ground
141	IORD#	output	output_low	ISA I/O read (+ ATC-1415 U13 pin 4 to both IDE p25)
142	IOWR#	output	output_low	ISA I/O write (+ ATC-1415 U13 pin 2 to both IDE p23)
143	DACK3#	output	output_low	ISA DMA acknowledge 3
144	DACK1#	output	output_low	ISA DMA acknowledge 1
145	REFRESH#	bidir	inverted	I/O refresh control, can work as ISA refresh output
146	DACK2#	output	output_low	ISA DMA acknowledge 2
147	BALE	output	line	ISA bus address latch enable
148	OSC	input	clock	Oscilator 14.318 MHz input (ATC-1415 via 22R to U17 clock generator pin 4, other branch via 22R to ISA OSC)
149	IRQ8#	input	input_low	RTC interrupt
150	IRQ7	input	line	ISA interrupt 7
151	IRQ6	input	line	ISA interrupt 6
152	IRQ5	input	line	ISA interrupt 5
153	IRQ4	input	line	ISA interrupt 4
154	IRQ3	input	line	ISA interrupt 3
155	?IRQ1	input	line	?ISA interrupt 1 (TODO ATC-1415 seems to be not connected)
156	GND	power_in	line	Ground
157	RSTDRV	output	line	Reset to rest of the system (including ISA, excluding CPU)
158	IRQ9	input	line	ISA interrupt 9 (ATC-1415 ISA B4)
159	SD7	tri_state	line	ISA data 7
160	SD6	tri_state	line	ISA data 6
161	SD5	tri_state	line	ISA data 5
162	SD4	tri_state	line	ISA data 4
163	SD3	tri_state	line	ISA data 3
164	SD2	tri_state	line	ISA data 2
165	SD1	tri_state	line	ISA data 1
166	SD0	tri_state	line	ISA data 0
167	IOCHRDY#	bidirectional	inverted	ISA bus I/O channel ready (device insert wait states)
168	AEN	output	line	ISA address enable (DMA controls the bus)
169	SA19/LA19	tri_state	line	ISA address 19 / ISA latchable address 19
170	GND	power_in	line	Ground
171	GND	power_in	line	Ground
172	SA18/LA18	tri_state	line	ISA address 18 / ISA latchable address 18 (on biostar)

Pin	Name	Direction	Type	Description
173	SA17/LA17	tri_state	line	ISA address 17 / ISA latchable address 17 (on biostar)
174	SA16	tri_state	line	ISA address 16
175	SA15	tri_state	line	ISA address 15
176	SA14	tri_state	line	ISA address 14
177	SA13	tri_state	line	ISA address 13
178	SA12	tri_state	line	ISA address 12
179	SA11	tri_state	line	ISA address 11
180	SA10	tri_state	line	ISA address 10
181	SA9	tri_state	line	ISA address 9
182	VCC	power_in	line	Power supply +5V
183	SA8	tri_state	line	ISA address 8
184	SA7	tri_state	line	ISA address 7
185	SA6	tri_state	line	ISA address 6
186	SA5	tri_state	line	ISA address 5
187	SA4	tri_state	line	ISA address 4
188	SA3	tri_state	line	ISA address 3
189	SA2	tri_state	line	ISA address 2
190	SA1	tri_state	line	ISA address 1
191	SA0	tri_state	line	ISA address 0
192	SBHE#	tri_state	line	ISA high byte [15:8] enable
193	LA23	tri_state	line	ISA latchable address 23
194	GND	power_in	line	Ground
195	LA22	tri_state	line	ISA latchable address 22
196	LA21	tri_state	line	ISA latchable address 21
197	LA20	tri_state	line	ISA latchable address 20
198	LA19	tri_state	line	ISA latchable address 19 (ATC-1415) / (RN17.4 biostar, SA19 is used instead)
199	LA18/MS_DATA	tri_state	line	ISA latchable address 18 / Mouse data
200	LA17/MS_CLK	tri_state	line	ISA latchable address 17 / Mouse clock
201	SD8	tri_state	line	ISA data 8
202	SD9	tri_state	line	ISA data 9
203	SD10	tri_state	line	ISA data 10
204	SD11	tri_state	line	ISA data 11
205	SD12	tri_state	line	ISA data 12
206	SD13	tri_state	line	ISA data 13
207	SD14	tri_state	line	ISA data 14
208	VCC	power_in	line	Power supply +5V

IBC UM8886-N pin list overview

1	GND	53	PCICLKO	105	VCC	157	RSTDRV
2	SD15	54	SMI2#/LB2#/KBCLKI	106	IDSEL	158	IRQ9
3	MEMW#	55	KBCLK	107	CBE3#	159	SD7
4	MEMR#	56	PGP0_RC#	108	AD24	160	SD6
5	MEMCS16#	57	PGP1/LDEV#/GA20	109	AD25	161	SD5
6	IOCS16#	58	ROMCS#_KBCS#	110	AD26	162	SD4
7	IRQ10	59	SPKR	111	AD27	163	SD3
8	IRQ11	60	INTD#	112	AD28	164	SD2
9	IRQ12	61	INTC#	113	AD29	165	SD1
10	IRQ14	62	INTB#	114	AD30	166	SD0
11	IRQ15	63	INTA#	115	AD31	167	IOCHRDY#
12	GND	64	AD0	116	GND	168	AEN
13	DACK0#/MDACK2#	65	AD1	117	GNT0#/NCLKDC#	169	SA19
14	DACK5#/ENBUF#	66	AD2	118	GNT1#	170	GND
15	DACK6#/UNDOCKGNT	67	GND	119	GNT2#/SUSP#	171	GND
16	DACK7#/DKDACKA#/SCLK	68	GND	120	GNT3#/NCLKDC#	172	SA18
17	DREQ0/MDRQ2/MUXSEL2	69	AD3	121	CPUGNT#	173	SA17
18	DREQ1/MDRQ1/MUXSEL1	70	AD4	122	PWRLATCH2_MEMREQ	174	SA16
19	DREQ2/MUXIN	71	AD5	123	X32K_MEMACK	175	SA15
20	DREQ3/MDRQ0/MUXSEL0	72	AD6	124	CPUREQ#	176	SA14
21	DREQ5/DSPWROK	73	AD7	125	REQ0#/COVERSW#	177	SA13
22	DREQ6/CD#	74	CBE0#	126	REQ1#	178	SA12
23	DREQ7/DK_REQA#/SDATA1	75	AD8	127	REQ2#/891BUSY#/DK_REQB#	179	SA11
24	RSTIN#	76	AD9	128	REQ3#/COVERSW#	180	SA10
25	FERR#	77	PCICLKI	129	CLKIN	181	SA9
26	VCC3V	78	VCC	130	VCC	182	VCC
27	VCC	79	AD10	131	VCC	183	SA8
28	IGNNE#	80	AD11	132	EOP	184	SA7
29	CPURSTO	81	AD12	133	BCLK	185	SA6
30	NMI	82	AD13	134	IDEHDEN#	186	SA5
31	A20M#	83	AD14	135	MASTER#	187	SA4
32	SMI#	84	AD15	136	IOCHCK#/CLKRUN#	188	SA3
33	STCK#/CYRIXSUSP#	85	CBE1#	137	0WS#/ACIN	189	SA2
34	PWRLATCH1	86	PAR	138	SMEMR#	190	SA1
35	EXTSMI1#_EPMI	87	SERR#	139	SMEMW#	191	SA0
36	IDE17X#/PGP3#	88	LOCK#	140	GND	192	SBHE#
37	INTR	89	STOP#	141	IORD#	193	LA23
38	PTEST#	90	GND	142	IOWR#	194	GND
39	GND	91	DEVSEL#	143	DACK3#/MDACK0#	195	LA22
40	XDEN/PGP2#/AS01	92	TRDY#	144	DACK1#/MDACK1#	196	LA21
41	XDIR/PGP3#/TCWR	93	IRDY#	145	REFRESH#	197	LA20
42	RTCAS	94	FRAME#	146	DACK2#/CD1#	198	LA19
43	RTCWR#	95	CBE2#	147	BALE	199	LA18
44	RTCRD#	96	AD16	148	OSC	200	LA17
45	IDE1FX#_AS0#	97	AD17	149	IRQ8#	201	SD8
46	IDE3FX#_AS1#	98	AD18	150	IRQ7	202	SD9
47	ONOFF#/ACIN	99	AD19	151	IRQ6	203	SD10
48	LB1#/LB2#	100	AD20	152	IRQ5	204	SD11
49	IDE37X#/PGP2#	101	AD21	153	IRQ4	205	SD12
50	SMIACT#	102	AD22	154	IRQ3	206	SD13
51	HCLKO	103	AD23	155	IRQ1	207	SD14
52	GND	104	GND	156	GND	208	VCC

IBC UM8886-N pin description

No pin tracing was done for this IBC variant.

Pin	Name	Direction	Type	Description
1	GND	power_in	line	Ground
2	SD15	tri_state	line	ISA data 15
3	MEMW#	tri_state	inverted	ISA memory write
4	MEMR#	tri_state	inverted	ISA memory read
5	MEMCS16#	tri_state	inverted	ISA 16bit memory chip select
6	IOCS16#	input	input_low	ISA 16bit I/O chip select
7	IRQ10	input	line	ISA interrupt 10
8	IRQ11	input	line	ISA interrupt 11
9	IRQ12	input	line	ISA interrupt 12
10	IRQ14	input	line	ISA interrupt 14
11	IRQ15	input	line	ISA interrupt 15
12	GND	power_in	line	Ground
13	DACK0#/MDACK2#	bidir	inverted	ISA DMA acknowledge 0 / PNP,dock,expand mode?
14	DACK5#/ENBUF#	bidir	inverted	ISA DMA acknowledge 5 / Output enable docking signal buffers
15	DACK6#/UNDOCKGN	bidir	inverted	ISA DMA acknowledge 6 / Undocking granted
16	DACK7#/DKDACKA#	output	inverted	ISA DMA acknowledge 7 / Dock DMA acknowledge / ?Serial clock
17	DREQ0/MDRQ2/MUX	input	line	ISA DMA request 0 / PNP-dock DMA request bit 2 / External DMA multiplexer select bit 2
18	DREQ1/MDRQ1/MUX	input	line	ISA DMA request 1 / PNP-dock DMA request bit 1 / External DMA multiplexer select bit 1
19	DREQ2/MUXIN	input	line	ISA DMA request 2 (PNP-dock DMA from floppy) / Input from external DMA multiplexer output
20	DREQ3/MDRQ0/MUX	input	line	ISA DMA request 3 / PNP-dock DMA request bit 0 / External DMA multiplexer select bit 0
21	DREQ5/DSPWROK	input	line	ISA DMA request 5 / Docking station power OK
22	DREQ6/CD#	input	line	ISA DMA request 6 / Dock-on or card detected
23	DREQ7/DK_REQA#/S	input	line	ISA DMA request 7 / PCI bus request from docking station / ???
24	RSTIN#	input	input_low	Reset input from power good circuitry
25	FERR#	input	input_low	FPU error
26	VCC3V	power_in	line	Power supply +3.3V
27	VCC	power_in	line	Power supply +5V
28	IGNNE#	output	output_low	Ignore FPU error
29	CPURSTO	output	line	CPU reset
30	NMI	output	line	Nonmaskable interrupt to CPU
31	A20M#	output	output_low	Mask CPU address 20
32	SMI#	bidirectional	inverted	System management interrupt, output request, input SMM status
33	STCK#/CYRIXSUSP#	output	output_low	CPU stop clock / ?Cyrix suspend
34	PWRLATCH1	output	line	Latch signal to generate PM signals (C6,C7/SUSP,C8,C9,CLKSEL[2:0]) from ISA data
35	EXTSMI1#_EPMI	input	input_low	External SMI 1
36	IDE17X#/PGP3#	output	output_low	IDE 17Xh chip select / Programmable output pin 3
37	INTR	output	line	Maskable CPU interrupt
38	PTEST#	input	input_low	Test pin, pull high?
39	GND	power_in	line	Ground

Pin	Name	Direction	Type	Description
40	XDEN/PGP2#/AS01	output	output_low	XD bus (8bit ISA) data buffer enable / Programmable output pin 3 / ???
41	XDIR/PGP3#/TCWR	output	line	XD bus (8bit ISA) data buffer direction / Programmable output pin 2 / RTC 4K RAM R/W signal
42	RTCAS	output	line	RTC address latch
43	RTCWR#	output	output_low	RTC write
44	RTCRD#	output	output_low	RTC read
45	IDE1FX#_AS0#	output	output_low	IDE 1FXh chip select
46	IDE3FX#_AS1#	output	output_low	IDE 3FXh chip select
47	ONOFF#/ACIN	input	input_low	On/Off switch / AC in (detection?)
48	LB1#/LB2#	input	input_low	Low battery 1 / Low battery 2
49	IDE37X#/PGP2#	output	output_low	IDE 37Xh chip select / Programmable output pin 2
50	SMIACT#	input	input_low	System management interrupt acknowledge from CPU
51	HCLKO	output	clock	Global clock output to CPU and other parts of the chipset (50/60/66 MHz)
52	GND	power_in	line	Ground
53	PCICLKO	output	clock	Global clock output to the PCI subsystem
54	SMI2#/LB2#/KBCLKI	input	input_low	External SMI 2 / Low battery 2 / Keyboard clock (input?)
55	KBCLK	output	line	Keyboard clock (?output to 8042)
56	PGP0_RC#	bidirectional	line	Reset input from 8042, programmable input/output pin 0
57	PGP1/LDEV#/GA20	bidirectional	inverted	Programmable input/ouput pin 1 / VLB local device / A20 gate from 8042
58	ROMCS#_KBCS#	output	output_low	ROM chip select for memory cycles, keyboard controller chip select for I/O port cycles
59	SPKR	output	line	Speaker output
60	INTD#	input	input_low	PCI interrupt D
61	INTC#	input	input_low	PCI interrupt C
62	INTB#	input	input_low	PCI interrupt B
63	INTA#	input	input_low	PCI interrupt A
64	AD0	tri_state	line	PCI Address/Data 0
65	AD1	tri_state	line	PCI Address/Data 1
66	AD2	tri_state	line	PCI Address/Data 2
67	GND	power_in	line	Ground
68	GND	power_in	line	Ground
69	AD3	tri_state	line	PCI Address/Data 3
70	AD4	tri_state	line	PCI Address/Data 4
71	AD5	tri_state	line	PCI Address/Data 5
72	AD6	tri_state	line	PCI Address/Data 6
73	AD7	tri_state	line	PCI Address/Data 7
74	CBE0#	tri_state	inverted	PCI command/byte enable 0
75	AD8	tri_state	line	PCI Address/Data 8
76	AD9	tri_state	line	PCI Address/Data 9
77	PCICLKI	input	clock	PCI clock input
78	VCC	power_in	line	Power supply +5V
79	AD10	tri_state	line	PCI Address/Data 10
80	AD11	tri_state	line	PCI Address/Data 11
81	AD12	tri_state	line	PCI Address/Data 12
82	AD13	tri_state	line	PCI Address/Data 13
83	AD14	tri_state	line	PCI Address/Data 14

Pin	Name	Direction	Type	Description
84	AD15	tri_state	line	PCI Address/Data 15
85	CBE1#	tri_state	inverted	PCI command/byte enable 1
86	PAR	tri_state	line	PCI parity bit
87	SERR#	input	inverted	PCI system error, will generate NMI
88	LOCK#	input	inverted	PCI lock signal
89	STOP#	tri_state	inverted	PCI stop signal
90	GND	power_in	line	Ground
91	DEVSEL#	tri_state	inverted	PCI device select
92	TRDY#	tri_state	inverted	PCI target ready
93	IRDY#	tri_state	inverted	PCI Initiator ready
94	FRAME#	tri_state	inverted	PCI frame signal
95	CBE2#	tri_state	inverted	PCI command/byte enable 2
96	AD16	tri_state	line	PCI Address/Data 16
97	AD17	tri_state	line	PCI Address/Data 17
98	AD18	tri_state	line	PCI Address/Data 18
99	AD19	tri_state	line	PCI Address/Data 19
100	AD20	tri_state	line	PCI Address/Data 20
101	AD21	tri_state	line	PCI Address/Data 21
102	AD22	tri_state	line	PCI Address/Data 22
103	AD23	tri_state	line	PCI Address/Data 23
104	GND	power_in	line	Ground
105	VCC	power_in	line	Power supply +5V
106	IDSEL	input	line	PCI ID select from northbridge, connects to AD18
107	CBE3#	tri_state	inverted	PCI command/byte enable 3
108	AD24	tri_state	line	PCI Address/Data 24
109	AD25	tri_state	line	PCI Address/Data 25
110	AD26	tri_state	line	PCI Address/Data 26
111	AD27	tri_state	line	PCI Address/Data 27
112	AD28	tri_state	line	PCI Address/Data 28
113	AD29	tri_state	line	PCI Address/Data 29
114	AD30	tri_state	line	PCI Address/Data 30
115	AD31	tri_state	line	PCI Address/Data 31
116	GND	power_in	line	Ground
117	GNT0#/NCLKDC#	output	output_low	PCI grant to master 0 / CPU clock will be D.C.
118	GNT1#	output	output_low	PCI grant to master 1
119	GNT2#/SUSP#	output	output_low	PCI grant to master 2 / Suspend request to CPU / Dock DMA acknowledge B / Serial data 3 to dock
120	GNT3#/NCLKDC#	output	output_low	PCI grant to master 3 / CPU clock will be D.C.
121	CPUGNT#	output	output_low	PCI CPU grant
122	PWRLATCH2_MEMRE	output	line	Latch signal to generate PM signals (C0-C5, C8, CKGENPD) from ISA data
123	X32K_MEMACK	input	line	RTC 32768 Hz clock
124	CPUREQ#	input	input_low	PCI request from CPU
125	REQ0#/COVERSW#	input	input_low	PCI request from master 0 / LCD cover has been closed
126	REQ1#	input	input_low	PCI request from master 1
127	REQ2#/891BUSY#/D	input	input_low	PCI request from master 2 / ??? / Dock request
128	REQ3#/COVERSW#	input	input_low	PCI request from master 3 / LCD cover has been closed
129	CLKIN	input	clock	Main clock input from generator (50/60/66/etc. MHz), ?generates FSB/PCI from this

Pin	Name	Direction	Type	Description
130	VCC	power_in	line	Power supply +5V
131	VCC	power_in	line	Power supply +5V
132	EOP	bidirectional	line	End of DMA process
133	BCLK	output	clock	ISA SYSCLK
134	IDEHDEN#	output	output_low	IDE cycle enable (for ISA/IDE buffer)
135	MASTER#	input	input_low	ISA 16bit master drives bus
136	IOCHCK#/CLKRUN#	input	input_low	ISA I/O channel check (will generate NMI) / reserved
137	0WS#/ACIN	input	input_low	Zero wait states request from ISA / ? AC plugged in?
138	SMEMR#	output	output_low	ISA system memory read (B12)
139	SMEMW#	output	output_low	ISA system memory write (B11)
140	GND	power_in	line	Ground
141	IORD#	output	output_low	ISA I/O read
142	IOWR#	output	output_low	ISA I/O write
143	DACK3#/MDACK0#	output	output_low	ISA DMA acknowledge 3 / PNP-dock DACK
144	DACK1#/MDACK1#	output	output_low	ISA DMA acknowledge 1 / PNP-dock DACK
145	REFRESH#	bidir	inverted	I/O refresh control, can work as ISA refresh output
146	DACK2#/CD1#	bidir	inverted	ISA DMA acknowledge 2 / card detect 1
147	BALE	output	line	ISA bus address latch enable
148	OSC	input	clock	Oscilator 14.318 MHz input
149	IRQ8#	input	input_low	RTC interrupt
150	IRQ7	input	line	ISA interrupt 7
151	IRQ6	input	line	ISA interrupt 6
152	IRQ5	input	line	ISA interrupt 5
153	IRQ4	input	line	ISA interrupt 4
154	IRQ3	input	line	ISA interrupt 3
155	IRQ1	input	line	ISA interrupt 1
156	GND	power_in	line	Ground
157	RSTDRV	output	line	Reset to rest of the system (excluding CPU)
158	IRQ9	input	line	ISA interrupt 9
159	SD7	tri_state	line	ISA data 7
160	SD6	tri_state	line	ISA data 6
161	SD5	tri_state	line	ISA data 5
162	SD4	tri_state	line	ISA data 4
163	SD3	tri_state	line	ISA data 3
164	SD2	tri_state	line	ISA data 2
165	SD1	tri_state	line	ISA data 1
166	SD0	tri_state	line	ISA data 0
167	IOCHRDY#	bidirectional	inverted	ISA bus I/O channel ready (device insert wait states)
168	AEN	output	line	ISA address enable (DMA controls the bus)
169	SA19	tri_state	line	ISA address 19
170	GND	power_in	line	Ground
171	GND	power_in	line	Ground
172	SA18	tri_state	line	ISA address 18
173	SA17	tri_state	line	ISA address 17
174	SA16	tri_state	line	ISA address 16
175	SA15	tri_state	line	ISA address 15
176	SA14	tri_state	line	ISA address 14
177	SA13	tri_state	line	ISA address 13
178	SA12	tri_state	line	ISA address 12
179	SA11	tri_state	line	ISA address 11
180	SA10	tri_state	line	ISA address 10

Pin	Name	Direction	Type	Description
181	SA9	tri_state	line	ISA address 9
182	VCC	power_in	line	Power supply +5V
183	SA8	tri_state	line	ISA address 8
184	SA7	tri_state	line	ISA address 7
185	SA6	tri_state	line	ISA address 6
186	SA5	tri_state	line	ISA address 5
187	SA4	tri_state	line	ISA address 4
188	SA3	tri_state	line	ISA address 3
189	SA2	tri_state	line	ISA address 2
190	SA1	tri_state	line	ISA address 1
191	SA0	tri_state	line	ISA address 0
192	SBHE#	tri_state	line	ISA high byte [15:8] enable
193	LA23	tri_state	line	ISA latchable address 23
194	GND	power_in	line	Ground
195	LA22	tri_state	line	ISA latchable address 22
196	LA21	tri_state	line	ISA latchable address 21
197	LA20	tri_state	line	ISA latchable address 20
198	LA19	tri_state	line	ISA latchable address 19
199	LA18	tri_state	line	ISA latchable address 18
200	LA17	tri_state	line	ISA latchable address 17
201	SD8	tri_state	line	ISA data 8
202	SD9	tri_state	line	ISA data 9
203	SD10	tri_state	line	ISA data 10
204	SD11	tri_state	line	ISA data 11
205	SD12	tri_state	line	ISA data 12
206	SD13	tri_state	line	ISA data 13
207	SD14	tri_state	line	ISA data 14
208	VCC	power_in	line	Power supply +5V

IBC registers

Register power-on values (UM8886-BF variant):

```
40: 01 04 08 9a | bc 00 10 30 | 00 00 00 00 | 00 00 00 00
50: 00 02 00 00 | 00 00 00 00 | 00 00 00 00 | 00 00 00 00
rest is zeroed
```

Register description:

```
40.10   Set on B2 revision, but cleared if no PS/2 mouse
40.04   PCI posted memory write
40.03   IBC (ISA bridge controller) devsel decoding (medium/slow/fast/resvd?)
40.02   if set (0x14->0x16), linux freezes

41.40   Disable parity check (maybe PCI SERR#?)
41.20   Set during UM8886BF FIFO mode and restored to prior state afterward
41.04   PCI bus park option 0=enabled (alt "Enhance PCI performance")

43.F0   INTA target IRQ
43.0F   INTB target IRQ
        NOTICE resulting IRQ may collide if incorrectly shared (PS/2 mouse + PCI)
        probably level vs edge triggered

44.F0   INTC target IRQ
44.0F   INTD target IRQ
```

```
45.04   Set on B2 revision
45.01   Set on B2 revision, but cleared if no PS/2 mouse

46.80   PM IRQ (10/15)
46.40   PM interrupt method (SMI/IRQ)
46.10   Preempt PCI master option
46.08   INTD enabled
46.04   INTC enabled
46.02   INTB enabled
46.01   INTA enabled

47.40   Enable flash writes (0=enabled, 1=disabled). Relation with 57.4 unknown
47.08   INTD level triggered
47.04   INTC level triggered
47.02   INTB level triggered
47.01   INTA level triggered

48-4F   UNKNOWN Doesn't retain written value

50.80   Set if PCI video BIOS is installed
        ??? is set even with ISA VGA
50.01   Set by bootblock after filling segment E and F

51.FE   Size of RAM in units of 4M (rounded down, exception: 4M if rounding down results in 0M)
51.01   Set by bootblock after filling segment E and F

56.80   UNKNOWN
56.60   Cleared by boot block, under some circumstances set by video card setup
56.0C   KBD clock (7MHz/by4/by3/by2)
56.03   ISA clock (by3/by4/by2/resvd?)

57.80   Enable segment C decode to ROM
        Will alias to segment E on 128kB flash
        Visible at FFFC0000
        Causes a bus conflict if ISA decodes too
57.40   Enable segment D decode to ROM
        Will alias to segment F on 128kB flash
        Visible at FFFD0000
        Causes a bus conflict if ISA decodes too
57.20   Enable segment E decode to ROM
        Visible at FFFE0000
        NOTICE segment F is always on
57.10   Set on B2 revision
57.08   Keyboard Emulation
57.04   Some bit used for flash write protection (Guessed: GPO, but seems wrong)
        8886AF/8886BF (Shuttle HOT433 and Biostar UUD8433)
            Set = Protected, Clear = Writeable
        8886F (Gigabyte GA486IM)
            Clear = Protected, Set = Writeable
57.03   IO recovery time (2BLCK/4BCLK/8BCLK/12BCLK)

58-6F   UNKNOWN Doesn't retain written value

70.80   Monitor PCI4 master activity
70.40   Monitor PCI3 master activity
70.20   Monitor PCI2 master activity
```

```
70.0F   Green Timer minutes (0.5/1/2/4/8/16/32/64/128/256/512/rsvd/rsvd/rsvd/disable/0.25)

71.80   Monitor PCI1 master activity
71.40   Monitor LPT access
71.20   Monitor COM access
71.10   Monitor ISA DMA master access
71.08   Monitor IDE access
71.04   Monitor Floppy access
71.02   Monitor Graphics card access

72.80   Monitor extra region A9
72.7E   Monitor extra region mask (1=don't care for A0..A5) Really A0?
72.01   Monitor VL slave access

73.FF   Monitor extra region A8-A1

74.01   Monitor ISA shared memory access (A0000-D0000)

76.40   Set in flash-related code
76.30   SMI interface mode (Intel/Cyrix classic/?/AMD classic)
76.08   Will be set by non-PCI VGA access? Cleared and probed in write-merge setup code
76.04   Cleared by APM init
76.02   Set by APM CPU idle, cleared by APM init

77-7F   UNKNOWN Doesn't retain written value

80.FF   unknown, accessible

81.FF   unknown, accessible

82.03   Set on B2 revision to 3

83.80   1: overloads system with spurious / irq 7
83.40   unknown, accessible, may freeze linux
83.7F   unknown

84-8F   UNKNOWN Doesn't retain written value

90.80   Wake-Up on IRQ7
90.40   Wake-Up on IRQ6
90.20   Wake-Up on IRQ5
90.10   Wake-Up on IRQ4
90.08   Wake-Up on IRQ3
90.04   maybe IRQ2 or gap (some IRQ are unusable)
90.02   maybe IRQ1 or gap (some IRQ are unusable)
90.01   maybe IRQ0 or gap (some IRQ are unusable)

91.80   Wake-Up on IRQ15
91.40   Wake-Up on IRQ14
91.20   maybe IRQ13 or gap (some IRQ are unusable)
91.10   Wake-Up on IRQ12
91.08   Wake-Up on IRQ11
91.04   Wake-Up on IRQ10
91.02   Wake-Up on IRQ9
91.01   Wake-Up on IRQ8
```

```
A0      Set to 0 by Boot Block
        Set to 34 by APM init (SMI mask?, 1 = enable)

A1      UNKNOWN
A1.20   if set, freezes linux until keyboard event o_O
A1.10   if set, freezes linux

A2.80   SMM event 7
A2.40   SMM event 6
A2.20   SMM event 5
A2.10   SMM event 4
A2.08   SMM event 3
A2.04   SMM event 2
A2.02   SMM event 1
A2.01   SMM event 0

A4.18   .18:  crash?
A4.03   CPU-to-PCI (2:1/1:1/3:2/resvd?)

A7.FF   unknown, seems to be set at random
        observed values 8f ef
        affected by IDE? 0->8 after disk access
        8f - ISA GPU, 0f - PCI GPU

A8.FF   Set to 0x28 on Biostar UUD8433
```

IBC IDE controller (only AF/BF/N chip variants)

There is a difference between AF and BF/N variants. The UM8886-AF IDE is just an ISA controller, whereas UM8886-BF or UM8886-N have IDE designed as an integrated PCI device. Both have some proprietary configuration settings.

Using the integrated IDE will block the usage of both the ISA and PCI buses (instead of a dedicated PCI IDE card), however the chipset is probably too simple to support any concurrent buses operations anyway.

It seems I/O ports are aliased at `0x400` granularity (`0x1f0`, `0x5f0`, `0x9f0`, ...) over the entire I/O port space (excluding decoded PCI BARs).

IBC UM8886-AF IDE controller

The UM8886-AF FIFO is 8 bytes in size.

The presumably same IDE controller design was also sold as a discrete UM8673-F chip on a PCI card but configuration set by jumpers, as discussed at UMC IDE/EIDE controller datasheets.

IBC UM8886-AF IDE config registers IBC has a register window located at I/O port `0x108`.

```
108     Index register
109     Read/write data register
```

Access needs to be unlocked by writing an unlock sequence. This sequence consists of writing a value `0x4a` followed by `0x6c` to index port. After that you can write following register indexes to index register and write/read data to/from the data register.

Access can then be locked again by writing a value of `0x34` to the index register.

```
B0.80   Enable primary IDE channel
B0.40   Enable secondary IDE channel
```

```
B2.F0   Something for primary master (0=fast, 4=slow)
B2.0F   Something for primary slave

B3.F0   Something for secondary master
B3.0F   Something for secondary slave

B4.F0   Another thing for primary master (3=fast, 12=slow)

B5.0F   Another thing for primary slave

B6.F0   Another thing for secondary master
B6.0F   Another thing for secondary slave

B7.F0   Third thing for primary master (2=fast, 12=slow)
B7.0F   Third thing for primary slave

B8.F0   Third thing for secondary master
B8.0F   Third thing for secondary slave
```

IBC UM8886-BF/N IDE PCI controller

The UM8886-BF FIFO is 60 bytes in size. The controller doesn't check sector size, multiple sector transfers, ATA vs. ATAPI commands, etc. so it has to be told when data transmission direction changes or finishes. A custom (nonstandard) IRQ 14/15 handler must check it and set the configuration from *filling* to *draining* mode at the correct moment (and do it for each transfer). This means only a specialised driver can use FIFO.

The UM8886-BF bus master does not support hardware scatter-gather, limiting its utility in a multi-tasking OS as the driver has to watch 16-bit register `0x58` count down to zero before loading next physical address.

There are also alignment restrictions leading to an unaligned head/tail that gets accessed in FIFO PIO or plain PIO mode when doing a bus master read.

IBC UM8886-BF/N IDE config registers

```
40.40   Set when FIFO is being accessed using bus mastering, clear otherwise
40.20   Set when FIFO is being accessed using PIO, clear otherwise
40.10   Set when FIFO is filling, clear when draining or disabled
40.08   Set when secondary interface owns FIFO, clear for primary or FIFO disabled

41.80   Enable primary channel
41.40   Enable secondary channel
41.0F   Unknown, set to 0D if north bridge is UM8881N/UM8891N or 09 otherwise
41.04   Cleared above PCI33

42.FF   Set to 33 on boot if north bridge is NexGen/UM8881N/UM8891N or 30 otherwise

43.C0   Timing A, primary master
        0: PIO3, PIO4 (fastest)
        1: PIO2
        2: PIO1
        3: PIO0 (slowest)
        other drives has the same table
43.30   Timing A, primary slave
43.0C   Timing A, secondary master
```

```
43.03   Timing A, secondary slave

44.F0   Timing B, primary master
        2: PIO4 (fastest)
        3: PIO3
        4: PIO2
        5: PIO1
        6: PIO0 (slowest)
        other drives has the same table
44.0F   Timing B, primary slave

45.F0   Timing B, secondary master
45.0F   Timing B, secondary slave

46.F0   Timing C, primary master
        0: PIO4 (fastest)
        1: PIO3
        5: PIO2
        8: PIO1
        b: PIO0 (slowest)
        other drives has the same table
46.0F   Timing C, primary slave

47.F0   Timing C, secondary master
47.0F   Timing C, secondary slave

48-4B   UNKNOWN, writeable, only 0x55555555 observed
        could be some timing base?
4C-4D   UNKNOWN, writeable, only 0x8888 observed
        could be some timing base?
4E-4F   UNKNOWN, writeable, only 0xAAAA observed
        could be some timing base?

50-53   UNKNOWN, writeable, only 0 observed

54-57   Bus mastering physical memory address

58-59   Bus mastering number of DWORDs remaining to transfer

5A-FF   unimplemented (write 0xff, returns 0x00)
```

Super I/O (SIO) UM82C863-F, UM8663-AF/BF

These are a wide UMC family of ISA SuperIO chips. The features varies with entire family. Usage spans between ISA/VLB I/O cards and 486 to Pentium motherboards. The descendants seems to be used in P6 generation machines too. Possible features are:

- Floppy drive controller
- Parallel port, optionally with ECP
- Two UART controllers, optionally with FIFO and IrDA
- Gameport decoder
- IDE control signals decoder
- Integrates a crystal oscillator 24 MHz (or can use an external source)

Compared with other SIOs, these are very limited in functionality and have only 2-5 configuration registers. The functionality can be pin strapped at power-on.

No official documentation is available.

SIO types and variants

There are at least two UM8663 variants with AF and BF suffixes. The direct predecessor of UM8663 seems to be UM82C863-F. There is a register description in INTERLOGIC INDUSTRIES ASC486 VER. C manual, which can be used to extrapolate the functionality to UM8663. The manual was found *after* disassembling the BIOS to discover UM8663 registers :-D.

The manual for PT-627 specifies multiple versions, which each uses a different SIO:

- PT-627A
  - UM8672/UM82C863/UM8667
  - EIDE + Standard I/O function
- PT-627B
  - UM8672/UM8663/UM8667
  - EIDE + FIFO UARTs + ECP/EPP/SPP
- PT-627C
  - UM8672/UM8668/UM8667
  - EIDE + FIFO UARTs + ECP/EPP/SPP + 2.88M FDD
- PT-627D
  - UM8672/UM8662/UM8667
  - EIDE + FIFO UARTs + standard LPT

It seems UM8668 is a further improved version. The UM8662 chip seems to have worse functionality than UM8663-AF/BF.

The motherboard PCChips/Hsin Tech M530 seems to use UM8663-BF or UM8670-F chips. With PCB routing around SIO practically the same. The UM8670-F will be most likely more modern design.

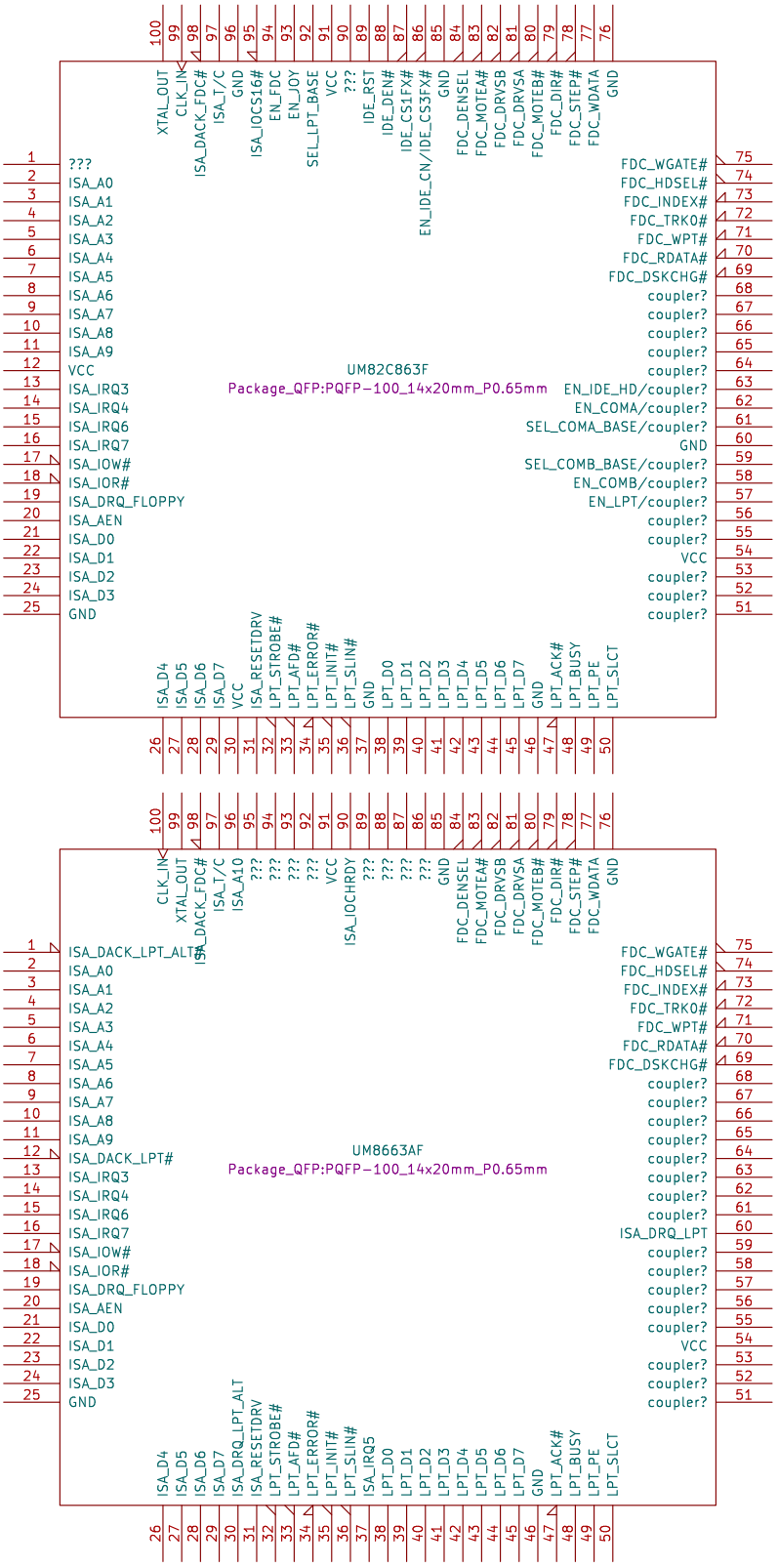
Some 486 boards carry UM8669-F, for which the relation with UM8663 is completely unknown.

The UM8663-AF and UM8663-BF may have a limited interchangeability. For example photos of Sowah Research SR-M401-A, PCChips/Hsin Tech M919 VER 3.2 and PCChips/Hsin Tech M507 can use both.

UM8663-AF/BF and UM82C863-F is not easily interchangeable. The oscillator input and output seems to be switched in the packages.

SIO pin assignment

SIO schematic symbols



SIO UM82C863-F pin list overview

1	???
2	ISA_A0
3	ISA_A1
4	ISA_A2
5	ISA_A3
6	ISA_A4
7	ISA_A5
8	ISA_A6
9	ISA_A7
10	ISA_A8
11	ISA_A9
12	VCC
13	ISA_IRQ3
14	ISA_IRQ4
15	ISA_IRQ6
16	ISA_IRQ7
17	ISA_IOW#
18	ISA_IOR#
19	ISA_DRQ_FLOPPY
20	ISA_AEN
21	ISA_D0
22	ISA_D1
23	ISA_D2
24	ISA_D3
25	GND
26	ISA_D4
27	ISA_D5
28	ISA_D6
29	ISA_D7
30	VCC

31	ISA_RESETDRV
32	LPT_STROBE#
33	LPT_AFD#
34	LPT_ERROR#
35	LPT_INIT#
36	LPT_SLIN#
37	GND
38	LPT_D0
39	LPT_D1
40	LPT_D2
41	LPT_D3
42	LPT_D4
43	LPT_D5
44	LPT_D6
45	LPT_D7
46	GND
47	LPT_ACK#
48	LPT_BUSY
49	LPT_PE
50	LPT_SLCT

51	coupler?
52	coupler?
53	coupler?
54	VCC
55	coupler?
56	coupler?
57	EN_LPT/coupler?
58	EN_COMB/coupler?
59	SEL_COMB_BASE/coupler?
60	GND
61	SEL_COMA_BASE/coupler?
62	EN_COMA/coupler?
63	EN_IDE_HD/coupler?
64	coupler?
65	coupler?
66	coupler?
67	coupler?
68	coupler?
69	FDC_DSKCHG#
70	FDC_RDATA#
71	FDC_WPT#
72	FDC_TRK0#
73	FDC_INDEX#
74	FDC_HDSEL#
75	FDC_WGATE#
76	GND
77	FDC_WDATA
78	FDC_STEP#
79	FDC_DIR#
80	FDC_MOTEB#

81	FDC_DRVSA
82	FDC_DRVSB
83	FDC_MOTEA#
84	FDC_DENSEL
85	GND
86	EN_IDE_CN/IDE_CS3FX#
87	IDE_CS1FX#
88	IDE_DEN#
89	IDE_RST
90	???
91	VCC
92	SEL_LPT_BASE
93	EN_JOY
94	EN_FDC
95	ISA_IOCS16#
96	GND
97	ISA_T/C
98	ISA_DACK_FDC#
99	CLK_IN
100	XTAL_OUT

SIO UM82C863-F pin description

Pins were traced on PT-606 ISA card, so part's references are for this board.

Pin	Name	Direction	Type	Description
1	???	unspecified	line	Not connected on PT-606
2	ISA_A0	input	line	ISA address 0
3	ISA_A1	input	line	ISA address 1
4	ISA_A2	input	line	ISA address 2
5	ISA_A3	input	line	ISA address 3
6	ISA_A4	input	line	ISA address 4
7	ISA_A5	input	line	ISA address 5
8	ISA_A6	input	line	ISA address 6
9	ISA_A7	input	line	ISA address 7
10	ISA_A8	input	line	ISA address 8
11	ISA_A9	input	line	ISA address 9
12	VCC	power_in	line	Power supply +5V
13	ISA_IRQ3	output	line	Interrupt request 3 (COM2)
14	ISA_IRQ4	output	line	Interrupt request 4 (COM1)
15	ISA_IRQ6	output	line	Interrupt request 6 (FDC)
16	ISA_IRQ7	output	line	Interrupt request 7 (LPT)
17	ISA_IOW#	input	input_low	ISA I/O write
18	ISA_IOR#	input	input_low	ISA I/O read
19	ISA_DRQ_FLOPPY	output	line	DMA request 2 (FDC)
20	ISA_AEN	input	line	ISA address enable

Pin	Name	Direction	Type	Description
21	ISA_D0	tri_state	line	ISA D0
22	ISA_D1	tri_state	line	ISA D1
23	ISA_D2	tri_state	line	ISA D2
24	ISA_D3	tri_state	line	ISA D3
25	GND	power_in	line	Ground
26	ISA_D4	tri_state	line	ISA D4
27	ISA_D5	tri_state	line	ISA D5
28	ISA_D6	tri_state	line	ISA D6
29	ISA_D7	tri_state	line	ISA D7
30	VCC	power_in	line	Power supply +5V?
31	ISA_RESETDRV	input	line	ISA reset
32	LPT_STROBE#	output	output_low	LPT strobe
33	LPT_AFD#	output	output_low	LPT autofeed
34	LPT_ERROR#	input	input_low	LPT error
35	LPT_INIT#	output	output_low	LPT init
36	LPT_SLIN#	output	output_low	LPT select printer
37	GND	power_in	line	Ground
38	LPT_D0	bidirectional	line	LPT data 0
39	LPT_D1	bidirectional	line	LPT data 1
40	LPT_D2	bidirectional	line	LPT data 2
41	LPT_D3	bidirectional	line	LPT data 3
42	LPT_D4	bidirectional	line	LPT data 4
43	LPT_D5	bidirectional	line	LPT data 5
44	LPT_D6	bidirectional	line	LPT data 6
45	LPT_D7	bidirectional	line	LPT data 7
46	GND	power_in	line	Ground
47	LPT_ACK#	input	input_low	LPT acknowledge
48	LPT_BUSY	input	line	LPT is busy
49	LPT_PE	input	line	LPT paper error
50	LPT_SLCT	input	line	LPT printer is selected
51	coupler?	???	line	UM82C865F.61
52	coupler?	???	line	UM82C865F.60
53	coupler?	???	line	UM82C865F.59
54	VCC	power_in	line	Supply +5V
55	coupler?	???	line	UM82C865F.58
56	coupler?	???	line	UM82C865F.57
57	EN_LPT/coupler?	???	line	Enable LPT strap + signal to UM82C865F.56
58	EN_COMB/coupler?	???	line	Enable COM port B strap + signal to UM82C865F.55
59	SEL_COMB_BASE/cou	???	line	Select COM port B as COM2/4 strap + signal to UM82C865F.54
60	GND	power_in	line	Ground
61	SEL_COMA_BASE/cou	???	line	Select COM port A as COM1/3 strap + signal to UM82C865F.53
62	EN_COMA/coupler?	???	line	Enable COM port A strap + signal to UM82C865F.52
63	EN_IDE_HD/coupler?	???	line	Enable IDE harddrive strap (?) + signal to UM82C865F.51
64	coupler?	???	line	signal to UM82C865F.50
65	coupler?	???	line	signal to UM82C865F.48
66	coupler?	???	line	signal to UM82C865F.47
67	coupler?	???	line	signal to UM82C865F.46
68	coupler?	???	line	signal to UM82C865F.45
69	FDC_DSKCHG#	input	input_low	FDC disk change
70	FDC_RDATA#	input	input_low	FDC read data

Pin	Name	Direction	Type	Description
71	FDC_WPT#	input	input_low	FDC write protect
72	FDC_TRK0#	input	input_low	FDC track 0
73	FDC_INDEX#	input	input_low	FDC index
74	FDC_HDSEL#	open_collector	output_low	FDC head select
75	FDC_WGATE#	open_collector	output_low	FDC write gate
76	GND	power_in	line	Ground
77	FDC_WDATA	open_collector	line	FDC write data
78	FDC_STEP#	open_collector	output_low	FDC head step/seek
79	FDC_DIR#	open_collector	output_low	FDC seek direction
80	FDC_MOTEB#	open_collector	output_low	FDC motor B enable
81	FDC_DRVSA	open_collector	output_low	FDC drive select A
82	FDC_DRVSB	open_collector	output_low	FDC drive select B
83	FDC_MOTEA#	open_collector	output_low	FDC motor A enable
84	FDC_DENSEL	open_collector	output_low	FDC density select
85	GND	power_in	line	Ground
86	EN_IDE_CN/IDE_CS3F	output	output_low	Enable IDE controller strap(?) + IDE chip select 3Fxh
87	IDE_CS1FX#	output	output_low	IDE chip select 1Fxh
88	IDE_DEN#	???	line	??? pull up to Vcc on PT-606
89	IDE_RST	output	line	IDE reset
90	???	???	line	Not connected on PT-606
91	VCC	power_in	line	Supply +5V
92	SEL_LPT_BASE	input	line	Choose LPT1/LPT2 base strap
93	EN_JOY	input	line	Gameport enable strap
94	EN_FDC	input	line	Floppy controller enable strap
95	ISA_IOCS16#	input	input_low	ISA I/O CS16
96	GND	power_in	line	Ground?
97	ISA_T/C	output	line	ISA T/C (end of DMA transfer)
98	ISA_DACK_FDC#	input	input_low	ISA DMA acknowledge (FDC, 2)
99	CLK_IN	input	clock	clock in from UM82C865F.64 or xtal in
100	XTAL_OUT	output	line	PT-606: N/C, ACE C MEGA-VL: xtal/clock out?

SIO UM8663-AF pin list overview

1	ISA_DACK_LPT_ALT#
2	ISA_A0
3	ISA_A1
4	ISA_A2
5	ISA_A3
6	ISA_A4
7	ISA_A5
8	ISA_A6
9	ISA_A7
10	ISA_A8
11	ISA_A9
12	ISA_DACK_LPT#
13	ISA_IRQ3
14	ISA_IRQ4
15	ISA_IRQ6
16	ISA_IRQ7
17	ISA_IOW#
18	ISA_IOR#
19	ISA_DRQ_FLOPPY
20	ISA_AEN
21	ISA_D0
22	ISA_D1
23	ISA_D2
24	ISA_D3
25	GND
26	ISA_D4
27	ISA_D5
28	ISA_D6
29	ISA_D7
30	ISA_DRQ_LPT_ALT

31	ISA_RESETDRV
32	LPT_STROBE#
33	LPT_AFD#
34	LPT_ERROR#
35	LPT_INIT#
36	LPT_SLIN#
37	ISA_IRQ5
38	LPT_D0
39	LPT_D1
40	LPT_D2
41	LPT_D3
42	LPT_D4
43	LPT_D5
44	LPT_D6
45	LPT_D7
46	GND
47	LPT_ACK#
48	LPT_BUSY
49	LPT_PE
50	LPT_SLCT

51	coupler?
52	coupler?
53	coupler?
54	VCC
55	coupler?
56	coupler?
57	coupler?
58	coupler?
59	coupler?
60	ISA_DRQ_LPT
61	coupler?
62	coupler?
63	coupler?
64	coupler?
65	coupler?
66	coupler?
67	coupler?
68	coupler?
69	FDC_DSKCHG#
70	FDC_RDATA#
71	FDC_WPT#
72	FDC_TRK0#
73	FDC_INDEX#
74	FDC_HDSEL#
75	FDC_WGATE#
76	GND
77	FDC_WDATA
78	FDC_STEP#
79	FDC_DIR#
80	FDC_MOTEB#

81	FDC_DRVSA
82	FDC_DRVSB
83	FDC_MOTEA#
84	FDC_DENSEL
85	GND
86	???
87	???
88	???
89	???
90	ISA_IOCHRDY
91	VCC
92	???
93	???
94	???
95	???
96	ISA_A10
97	ISA_T/C
98	ISA_DACK_FDC#
99	XTAL_OUT
100	CLK_IN

SIO UM8663-AF pin description

Pins were traced on Biostar motherboard and combined with UM82C863-F description.

It is highly likely the IDE signals use the same pins as UM82C863-F. There was no HW to trace that from.

Interesting fact: power supplies are routed through a diodes.

Pin	Name	Direction	Type	Description
1	ISA_DACK_LPT_ALT#	input	input_low	DMA acknowledge (1/3) for LPT2? (via R38 on biostar)
2	ISA_A0	input	line	ISA address 0
3	ISA_A1	input	line	ISA address 1
4	ISA_A2	input	line	ISA address 2
5	ISA_A3	input	line	ISA address 3
6	ISA_A4	input	line	ISA address 4
7	ISA_A5	input	line	ISA address 5
8	ISA_A6	input	line	ISA address 6
9	ISA_A7	input	line	ISA address 7
10	ISA_A8	input	line	ISA address 8

Pin	Name	Direction	Type	Description
11	ISA_A9	input	line	ISA address 9
12	ISA_DACK_LPT#	input	input_low	DMA acknowledge (1/3) for LPT
13	ISA_IRQ3	output	line	Interrupt request 3 (COM2)
14	ISA_IRQ4	output	line	Interrupt request 4 (COM1)
15	ISA_IRQ6	output	line	Interrupt request 6 (FDC)
16	ISA_IRQ7	output	line	Interrupt request 7 (LPT1)
17	ISA_IOW#	input	input_low	ISA I/O write
18	ISA_IOR#	input	input_low	ISA I/O read
19	ISA_DRQ_FLOPPY	output	line	DMA request 2 (FDC)
20	ISA_AEN	input	line	ISA address enable
21	ISA_D0	tri_state	line	ISA D0
22	ISA_D1	tri_state	line	ISA D1
23	ISA_D2	tri_state	line	ISA D2
24	ISA_D3	tri_state	line	ISA D3
25	GND	power_in	line	Ground
26	ISA_D4	tri_state	line	ISA D4
27	ISA_D5	tri_state	line	ISA D5
28	ISA_D6	tri_state	line	ISA D6
29	ISA_D7	tri_state	line	ISA D7
30	ISA_DRQ_LPT_ALT	output	line	DMA request 1 (if configured as LPT2?, biostar: trace under RTC)
31	ISA_RESETDRV	input	line	ISA reset
32	LPT_STROBE#	output	output_low	LPT strobe
33	LPT_AFD#	output	output_low	LPT autofeed
34	LPT_ERROR#	input	input_low	LPT error
35	LPT_INIT#	output	output_low	LPT init
36	LPT_SLIN#	output	output_low	LPT select printer
37	ISA_IRQ5	output	line	Interrupt request 5 (if configured as LPT2?, biostar: trace under ROM)
38	LPT_D0	bidirectional	line	LPT data 0
39	LPT_D1	bidirectional	line	LPT data 1
40	LPT_D2	bidirectional	line	LPT data 2
41	LPT_D3	bidirectional	line	LPT data 3
42	LPT_D4	bidirectional	line	LPT data 4
43	LPT_D5	bidirectional	line	LPT data 5
44	LPT_D6	bidirectional	line	LPT data 6
45	LPT_D7	bidirectional	line	LPT data 7
46	GND	power_in	line	Ground
47	LPT_ACK#	input	input_low	LPT acknowledge
48	LPT_BUSY	input	line	LPT is busy
49	LPT_PE	input	line	LPT paper error
50	LPT_SLCT	input	line	LPT printer is selected
51	coupler?	???	line	um8667.24
52	coupler?	???	line	um8667.23
53	coupler?	???	line	um8667.22
54	VCC	power_in	line	via diode to +5V
55	coupler?	???	line	um8667.21
56	coupler?	???	line	um8667.20
57	coupler?	???	line	um8667.18
58	coupler?	???	line	um8667.17
59	coupler?	???	line	um8667.16
60	ISA_DRQ_LPT	output	line	DMA request (1/3) for LPT
61	coupler?	???	line	um8667.15
62	coupler?	???	line	um8667.14
63	coupler?	???	line	um8667.13

Pin	Name	Direction	Type	Description
64	coupler?	???	line	um8667.12
65	coupler?	???	line	um8667.11
66	coupler?	???	line	um8667.10
67	coupler?	???	line	um8667.9
68	coupler?	???	line	um8667.8
69	FDC_DSKCHG#	input	input_low	FDC disk change
70	FDC_RDATA#	input	input_low	FDC read data
71	FDC_WPT#	input	input_low	FDC write protect
72	FDC_TRK0#	input	input_low	FDC track 0
73	FDC_INDEX#	input	input_low	FDC index
74	FDC_HDSEL#	open_collector	output_low	FDC head select
75	FDC_WGATE#	open_collector	output_low	FDC write gate
76	GND	power_in	line	Ground
77	FDC_WDATA	open_collector	line	FDC write data
78	FDC_STEP#	open_collector	output_low	FDC head step/seek
79	FDC_DIR#	open_collector	output_low	FDC seek direction
80	FDC_MOTEB#	open_collector	output_low	FDC motor B enable
81	FDC_DRVSA	open_collector	output_low	FDC drive select A
82	FDC_DRVSB	open_collector	output_low	FDC drive select B
83	FDC_MOTEA#	open_collector	output_low	FDC motor A enable
84	FDC_DENSEL	open_collector	output_low	FDC density select
85	GND	power_in	line	Ground
86	???	???	line	???
87	???	???	line	???
88	???	???	line	???
89	???	???	line	???
90	ISA_IOCHRDY	output	line	ISA I/O CHRDY
91	VCC	power_in	line	via diode to +5V
92	???	???		???
93	???	???		???
94	???	???		???
95	???	???		???
96	ISA_A10	input	line	ISA address 10
97	ISA_T/C	output	line	ISA T/C
98	ISA_DACK_FDC#	input	input_low	ISA DMA acknowledge (FDC, 2)
99	XTAL_OUT	output?	line	Xtal, most likely output
100	CLK_IN	input	clock	Clock in (biostar: via 33R to UM9515 pin 5)

SIO registers

The following access was found in BIOS for UM8663-AF variant:

```
108     Index register
109     Read/write data register
```

Similar to the IBC, the SuperIO access needs to be unlocked first. This is done by writing `0xaa` to the index port. Re-locking can be done by writing the value of `0x55`. It seems *any* access must be wrapped inside the lock/unlock sequence.

According to the manual the UM82C863-F implements `CR0` and `CR1` registers. The UM8663-AF has an additional register `CR2` and the UM8663-BF (untested) probably `CR3` and `CR4`. UM8663-BF variant probably supports IrDA on the serial port.

There are ISA I/O cards with UM82C863-F or UM8663-AF, where the configuration is set by jumpers. This means SIOs mounted on motherboards are using this mechanism too, most likely in a form of

preset pullup resistors. The pin value will be most likely sampled during the power-on and written into CRx registers, where it can be later changed via BIOS code.

Implemented registers and its power-on values:

Register	UM82C863-F	UM8663-AF (Biostar)	UM8663-BF
CR0	Untested	0x2f	Untested
CR1	Untested	0x9f	Untested
CR2	N/A	0x81	Untested
CR3	N/A	N/A	Untested
CR4	N/A	N/A	Untested

I/O base access of controllers is limited by the amount of used address signals. Which means there is an address aliasing at higher addresses (excluding PCI I/O space). The aliasing was checked for the LPT.

It seems BIOS access following configuration registers (CRx):

```
C0.80   UNKNOWN, some kind of status bit on UM8663-AF ?
        Reserved on UM82C863-F
C0.40   UNKNOWN, writeable to 1 on UM8663-AF
        Reserved on UM82C863-F
C0.20   Gameport enable
C0.10   IDE enable
C0.08   LPT enable
C0.04   COMB enable
C0.02   COMA enable
C0.01   FDC enable

C1.C0   LPT mode on UM8663-AF
        10: SPP
        01: EPP
        11: ECP
        00: disabled
  .40   LPT mode on UM82C863-F
        0: PS/2 compatible LPT (bidirectional?)
        1: AT mode compatible LPT
  .80   Test mode on UM82C863-F
        0: Test mode
        1: Normal mode
C1.20   UNTESTED, IDE existence?, defaults to 0
        0: already IDE in the system
        1: no IDE in the system
        NOTICE may have a different function on UM8663-AF
        TODO
            what is the meaning of this???
            maybe a complete IDE disable? (but already in C0.10)
C1.10   UNTESTED, IDE base
        0: 170  secondary
        1: 1f0  primary
C1.08   LPT base
        0: 278  LPT2
        1: 378  LPT1
C1.04   COMB base
        0: 2e8  COM4
        1: 2f8  COM2
```

```
C1.02   COMA base
        0: 3e8  COM3
        1: 3f8  COM1
C1.01   FDC base
        0: 370  secondary
        1: 3f0  primary

C2.80   Probably chip sleep, COM A/B, LPT, FDC become inoperative if cleared
        Set during FDC init
C2.40   UNKNOWN
C2.20   setting to 1 disables FDC
C2.10   UNKNOWN
C2.08   UNKNOWN
C2.04   Floppy 3-mode
        0: Disable
        1: Enable
C2.02   UNKNOWN
C2.01   UNKNOWN
        Set during FDC init

C3.FF   UNKNOWN, only UM8663-BF
        Additional LPT settings (probably IRQ, DMA)
        Additional COMA, COMB settings

C4.E0   UNKNOWN, only UM8663-BF
C4.10   UNTESTED UM8663-BF Serial port IR duplex mode
C4.0C   UNTESTED UM8663-BF Serial port used for IR
C4.03   UNTESTED UM8663-BF Serial port mode (Normal/HPSIR/ASKIR)
```

UM8667 I/O coupler

UM8667 is a complementary chip for UMC Super I/Os. It provides necessary level shifters and analog processing. Integrated functionality:

- Dual UART <-> RS232 level shifters for all signals
- Single gameport interface with A/D sampling (556 like) with output to a partial ISA interface

UM8667 I/O coupler predecessor

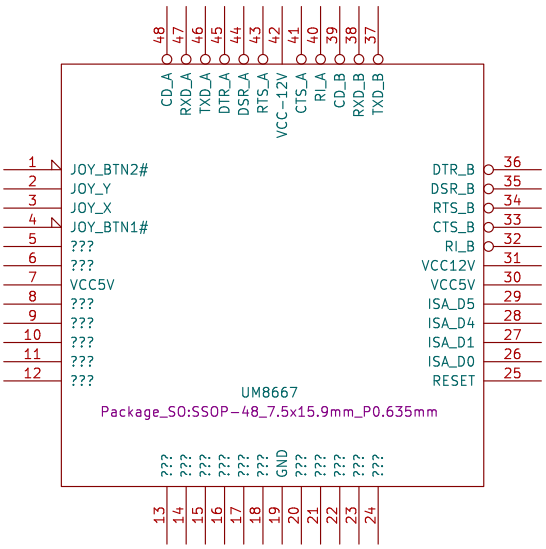
SIOs UM82C863F/UM8663 can be coupled with UM8667 or its predecessor UM82C865F. Both should have the same features, but UM82C865F includes a 24 MHz crystal oscillator so it can generate 24 MHz for SIO. It may be possible it has a support for dual joystick gameport. No further research has be made.

UM8667 I/O coupler pin assignment

As UM8667 implements a partial ISA bus (to access gameport controller), but it doesn't have ISA address. There needs to be at least IOR and IOW signals (or some equivalent) from SIO so I/O coupler can know when to read/write data (if gameport is enabled). Pins 5 and 6 are the candidates, but that's just a speculation (no HW to test on).

The TTL signals between RS232 level shifter and to/from SIO were not identified. It would require a RS232 voltage level injector/monitor to observe them (too much work :-D).

UM8667 I/O coupler schematic symbols



UM8667 I/O coupler pin list overview

1	JOY_BTN2#	25	RESET
2	JOY_Y	26	ISA_D0
3	JOY_X	27	ISA_D1
4	JOY_BTN1#	28	ISA_D4
5	???	29	ISA_D5
6	???	30	VCC5V
7	VCC5V	31	VCC12V
8	???	32	RI_B
9	???	33	CTS_B
10	???	34	RTS_B
11	???	35	DSR_B
12	???	36	DTR_B
13	???	37	TXD_B
14	???	38	RXD_B
15	???	39	CD_B
16	???	40	RI_A
17	???	41	CTS_A
18	???	42	VCC-12V
19	GND	43	RTS_A
20	???	44	DSR_A
21	???	45	DTR_A
22	???	46	TXD_A
23	???	47	RXD_A
24	???	48	CD_A

UM8667 I/O coupler pin description

Pins were traced on Biostar motherboard with help of PT-606 I/O ISA card, the descriptions may reference them.

Pin	Name	Direction	Type	Description
1	JOY_BTN2#	input	input_low	Joystick button 2 (biostar to +5V)
2	JOY_Y	input	line	Joystick analog axis Y
3	JOY_X	input	line	Joystick analog axis X
4	JOY_BTN1#	input	input_low	Joystick button 1 (biostar to +5V)
5	???	???		??? (biostar to +5V), possibly joystick control?
6	???	???		??? (biostar to +5V), possibly joystick control?
7	VCC5V	power_in	line	+5V supply
8	???	???		to um8663af.68
9	???	???		to um8663af.67
10	???	???		to um8663af.66
11	???	???		to um8663af.65
12	???	???		to um8663af.64
13	???	???		to um8663af.63
14	???	???		to um8663af.62
15	???	???		to um8663af.61
16	???	???		to um8663af.59
17	???	???		to um8663af.58
18	???	???		to um8663af.57
19	GND	power_in	line	Ground
20	???	???		to um8663af.56
21	???	???		to um8663af.55
22	???	???		to um8663af.53
23	???	???		to um8663af.52
24	???	???		to um8663af.51
25	RESET	input	line	reset
26	ISA_D0	tri_state	line	?ISA D0, possibly joystick
27	ISA_D1	tri_state	line	?ISA D1, possibly joystick
28	ISA_D4	tri_state	line	?ISA D4, possibly joystick
29	ISA_D5	tri_state	line	?ISA D5, possibly joystick
30	VCC5V	power_in	line	+5V supply
31	VCC12V	power_in	line	+12V via diode
32	RI_B	input	inverted	Port B Ring Indicator (RS232 pin 9)
33	CTS_B	input	inverted	Port B Clear To Send (RS232 pin 8)
34	RTS_B	output	inverted	Port B Request To Send (RS232 pin 7)
35	DSR_B	input	inverted	Port B Data Set Ready (RS232 pin 6)
36	DTR_B	output	inverted	Port B Data Terminal Ready (RS232 pin 4)
37	TXD_B	output	inverted	Port B TX data (RS232 pin 3)
38	RXD_B	input	inverted	Port B RX data (RS232 pin 2)
39	CD_B	input	inverted	Port B carrier detect (RS232 pin 1)
40	RI_A	input	inverted	Port A Ring Indicator (RS232 pin 9)
41	CTS_A	input	inverted	Port A Clear To Send (RS232 pin 8)
42	VCC-12V	power_in	line	-12V via diode
43	RTS_A	output	inverted	Port A Request To Send (RS232 pin 7)
44	DSR_A	input	inverted	Port A Data Set Ready (RS232 pin 6)
45	DTR_A	output	inverted	Port A Data Terminal Ready (RS232 pin 4)
46	TXD_A	output	inverted	Port A TX data (RS232 pin 3)
47	RXD_A	input	inverted	Port A RX data (RS232 pin 2)
48	CD_A	input	inverted	Port A carrier detect (RS232 pin 1)

UM8667 I/O coupler registers

There is only a single gameport register, but its I/O port `0x201` is decoded from SIO. UM8667 implements only ISA data bits 0, 1, 4 and 5, which corresponds to buttons and analog axes of the first joystick. The write to I/O port `0x201` should restart the conversion.

Code snippets

Fill RAM shadow segment E and F

```
HB  55.C0 = 00  Unlock F0000-FFFFF
... fill data ...
HB  54.01 = 1   E0000-EFFFF shadow read enable
IBC 51.01 = 1   Set after filling E/F
IBC 50.01 = 1   Set after filling E/F
HB  55.C0 = C0  Lock F0000-FFFFF
```